

Posterior thigh pain

with CARL ASKLING and ANTHONY SCHACHE

You know, this is a different hamstring—I did my left hamstring, I've done the right side of my back, I've just done my right hammy . . . obviously I've got injury concerns at the moment, now I have to go back and do what the experts tell me to give myself the best chance of being fully fit.

Michael Clarke, Australian cricket captain, December 2014

Pain in the 'hamstring region' can prove very frustrating for recreational and amateur athletes and may be career threatening for professional athletes. Hamstring muscle strains are the most common cause of posterior thigh pain, but referred pain to this area is also common. The average number of days until return to play (RTP) for hamstring injuries depends on the severity of the injury, but has been found to range between 18–60 days in elite level soccer players.¹

The incidence of recurrence is high. Up to one-third of hamstring injuries will recur, with the greatest risk being during the initial 2 weeks following return to sport.²

In this chapter, we focus on:

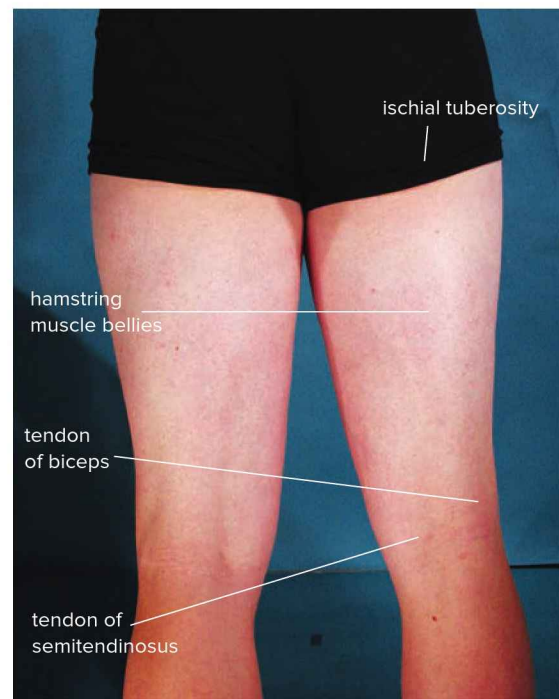
- relevant anatomy which is critical to diagnosis, prognosis and management
- clinical distinction between the major pathologies in the patient with posterior thigh pain
- the role of diagnostic imaging for this injury
- treatment approaches for the two types of acute hamstring injuries and for referred pain
- the indications for considering early or late surgical treatment
- how to make the often difficult return-to-play decision
- preventing the rightfully feared setback—hamstring strain recurrence.

FUNCTIONAL ANATOMY

The hamstring muscle group (Fig. 34.1) consists of three main muscles: biceps femoris, semimembranosus and semitendinosus. Biceps femoris has two heads: a long head and a short head. The long head is innervated by

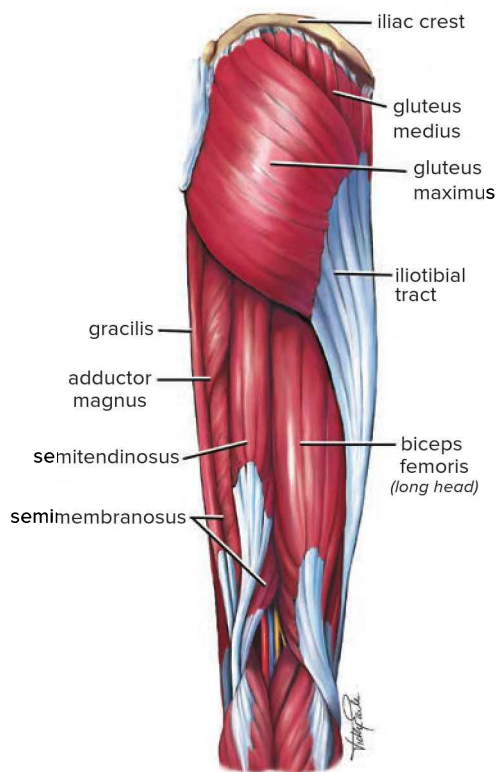
the tibial portion of the sciatic nerve (L5, S1–3), whereas the short head is innervated by the common peroneal portion (L5, S1–2).

The proximal hamstring complex (Fig. 34.1c) is a common site for pathology and has a complex anatomical arrangement.³

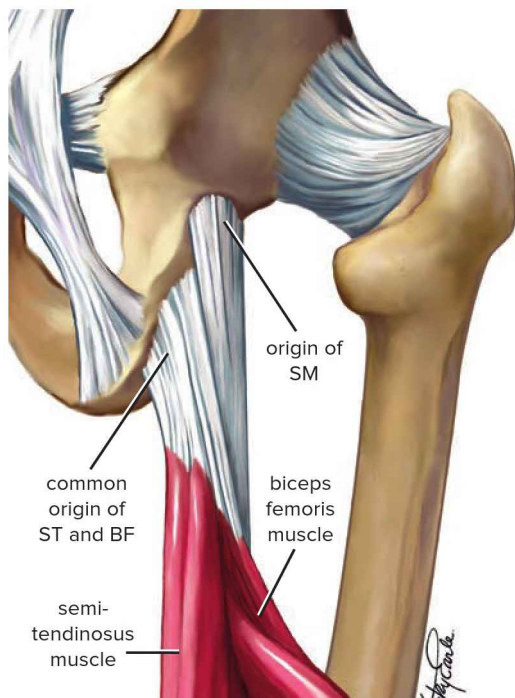


(a)

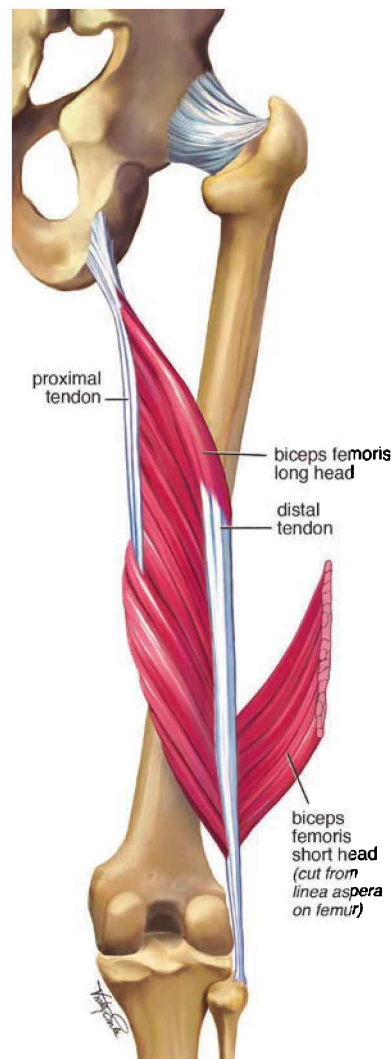
Figure 34.1 Anatomy of the posterior thigh (a) Surface anatomy



(b)



(c)



(d)

Figure 34.1 (cont.) (b) Muscles of the posterior thigh (c) Proximal hamstring origin (d) Anatomical arrangement of the long and short heads of biceps femoris

The long head of biceps femoris and semitendinosus share a common proximal tendon that arises from the medial facet of the ischial tuberosity. Semitendinosus muscle fibres originate from the ischial tuberosity and the medial aspect of the common tendon; the biceps femoris long head muscle fibres originate from the lateral aspect of the common tendon approximately 6 cm below the ischial tuberosity (Fig. 34.1d). The proximal free tendon of semimembranosus arises from the lateral facet of the ischial tuberosity and has a length of more than 10 cm. Moving distally, it extends medially passing ventral (deep)

to the semitendinosus/biceps femoris long head common proximal tendon.

The short head of biceps femoris arises from the linea aspera and thus only acts about the knee joint. Semitendinosus inserts onto the anteromedial surface of the proximal tibia (as part of the pes anserine muscle group), whereas semimembranosus inserts onto the medial tibial condyle. Biceps femoris long head and short head form a common distal tendon that has several insertions, including the lateral femoral condyle, fibula head, popliteus tendon and arcuate popliteal ligament.⁴

Both the proximal and distal tendons of the biceps femoris extend approximately 60% of the length of the muscle (Fig. 34.1d).⁵ As such, the biceps femoris muscle-tendon junction (a common site of injury) has a relatively unique arrangement that comprises two parts: the proximal muscle-tendon junction extending along the medial border and the distal muscle-tendon junction extending along the lateral border, with an area of overlap in the mid-zone of the muscle belly.

The ischiocondylar portion of the adductor magnus functions as if it was a 'hamstring' due to its anatomical alignment and innervation. Adductor magnus is a strong hip extensor muscle, especially when the hip is flexed. The ischiocondylar portion is innervated by the tibial portion of the sciatic nerve, like the majority of the hamstring group.

CLINICAL APPROACH

The key to effective management of posterior thigh pain is correct diagnosis. The clinician must determine whether the injury to the posterior thigh is an acute muscle injury (e.g. strain or contusion) or pain referred from elsewhere (e.g. lumbar spine). This differentiation can sometimes be challenging.

In healthy individuals, a strain to a large muscle group such as the hamstrings is the result of substantial force. The athlete will typically recall a particular point in time that the incident occurred and whether a significant force was applied to the muscle. The incident may be related to an eccentric contraction (e.g. sprinting) or it may be associated with an excessive stretch (e.g. ballet dancing).

Tethering of neural structures or fascial strains in the posterior thigh can also occur as an incident; however,

appropriate examination will reveal whether the injury has a neuromechanical or fascial component. Although the clinical examination may not present differently to a low-grade strain, the absence of evidence of muscle injury on magnetic resonance imaging (MRI) (10–20% of posterior thigh pain presentations) should make the clinician highly suspicious of a referred cause of pain.^{6–9}

The causes of posterior thigh pain are shown in Table 34.1.

History

Because there are different causes of posterior thigh pain, the clinician should use the history to develop a differential diagnosis that can then be refined further with a physical examination that is then appropriately structured.

Clinicians consider a wide range of factors in the history of the patient with posterior thigh pain. Some of these include the following.

1. Level of activity
 - a. Has a change in training coincided with injury?
 - b. Is there an adequate base of training?
2. Occupation/lifestyle
 - a. What factors outside of sport could be aggravating the condition (e.g. prolonged sitting at work, repetitive bending over with young children)?
3. Incident
 - a. Yes: consider strain, fascial/neural trauma.

If there was an incident, then what was the mechanism of injury? This information will help determine optimal rehabilitation exercises. Also, what was the severity of the pain experienced at the time of the injury? This information is important because a greater maximum pain score (visual analogue scale [VAS] 0–10) is associated with a longer time to RTP. For example, a VAS pain score of greater than 6 out of 10 has been shown to be significantly associated with an RTP time of >40 days.^{8, 10, 11}
 - b. No: consider overuse, referred pain, alternative abnormality.
4. Immediate impact on function
 - a. Being forced to stop participating in sport within 5 minutes of the initial incident is associated with an increased time to RTP.¹¹
5. Progress following injury
 - a. Slow: indicates a more severe injury
 - b. Erratic: strain is being aggravated by activity or injury is not a strain.
6. Ability to walk pain-free within one day following injury.

A study involving Australian rules football players found that those who could not walk pain-free within 1 day

PRACTICE PEARL

Clinicians should be reticent to diagnose an acute muscle strain in the absence of a convincing history of injury. Without a strong history of injury, consider referred pain.

Table 34.1 Causes of posterior thigh pain

Common	Less common	Not to be missed
Hamstring muscle strains • Type I • Type II • Recurrent Hamstring muscle contusion	Referred pain • Sacroiliac joint Tendinopathy • Biceps femoris • Semimembranosus/semitendinosus Bursitis • Semimembranosus • Ischiogluteal Fibrous adhesions 'Hamstring syndrome' (Chapter 30) Chronic compartment syndrome of the posterior thigh Apophysitis/avulsion fracture of the ischial tuberosity (in adolescents) Nerve entrapments • Posterior cutaneous thigh • Sciatic Adductor magnus strains Myositis ossificans, hamstring muscle	Tumours • Bone tumours Vascular • Iliac artery endofibrosis

were four times more likely to require >3 weeks to RTP compared to those who could.¹² Note, however, that a more recent study involving athletes from a variety of sports did not confirm this relationship.¹³

7. Aggravating factors

- Incident related: useful for specificity of rehabilitation (e.g. acceleration injuries require acceleration in the rehabilitation program)
- Non-incident related: eradication or modification for recovery and prevention (e.g. sitting at a computer causing back/hamstring pain requires ergonomic modification).

8. Behaviour with sport

- Warms up with activity, worse after: inflammatory pathology
- Starts with minimal or no pain, builds up with activity but not as severe after: claudicant, either neurological or vascular
- Sudden onset: mechanical (e.g. strain).

9. Night pain

- Sinister pathology
- Inflammatory condition.

10. Site of pain

- Posterior thigh and/or lower back: lumbar referral or neuromotor/biomechanical mediator
- Buttock, sacroiliac joint (SIJ) without lower back symptoms: gluteal trigger points
- Ischial: tendinopathy/bursitis/apophysitis/avulsion.

11. Presence of neurological symptoms

- Nerve involvement

12. Recurrent problem

- Extensive examination and rehabilitation required
- History of hamstring injury in the previous season is a strong predictor of risk of recurrence. For example, Australian Football players with a recent history of hamstring injury are 20 times more likely to suffer a recurrence within 3 weeks of RTP than those without a history.¹²

PRACTICE PEARL

Two items in the above list are particularly important:

- **3-the severity of the pain at the time of injury**
- **12-whether it is a recurrent problem.**

Physical examination

The examination further refines the distinction between local injury (i.e. acute muscle strain), referred pain or other unusual causes. A practical approach to assess various factors that commonly cause posterior thigh pain is outlined below.

1. Observation

- standing (Fig.34.2a)
- walking
- lying prone

2. Active movements

- lumbar movements
- hip extension
- standing leg swing
- supine active knee extension (Fig.34.2b)

3. Passive movements
 - a. hamstring muscle stretch (Fig.34.2c)
4. Resisted movements
 - a. knee flexion in isolation
 - b. hip extension in isolation
 - c. combined contraction—single leg bridge (Fig.34.2d)
5. Functional tests
 - a. running
 - b. kicking
 - c. sprint starts
6. Palpation
 - a. hamstring muscles (Fig.34.2e)
 - b. ischial tuberosity
 - c. gluteal muscles (Fig.34.2f)



(a)



(b)



(c)

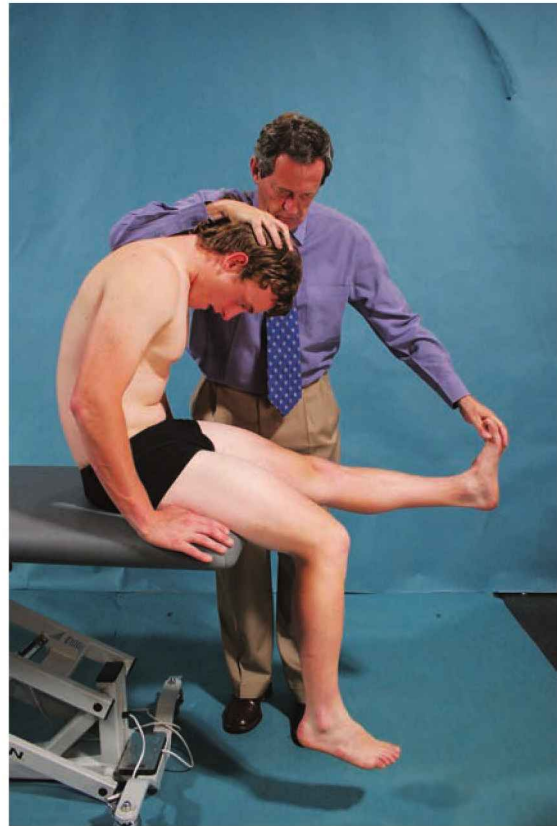


(d)

Figure 34.2 Examination of the patient with posterior thigh pain (a) Observation. Look for wasting, bruising or swelling of the posterior thigh. Observation of gait is also important. Observation of the lumbar spine may show the presence of an excessive lordosis or relative asymmetry. A lateral view may demonstrate excessive lumbar lordosis or anterior pelvic tilt (b) Active movement—active knee extension. The hip is actively flexed to 90° with the knee initially at 90° also. The knee is then slowly extended until pain is felt and then to the end of range (c) Passive movement—hamstring muscle stretch. The leg is raised to the point where pain is first felt and then to the end of range, pain permitting. Movement should be compared with the uninjured side (d) Combined contraction—single-leg bridge. A widely used ‘quick’ clinical assessment of resisted hamstring contraction is the single-leg bridge. This can be done with the knee fully extended or flexed to 90° (or any angle in between these two positions)



(e)



(g)



(f)

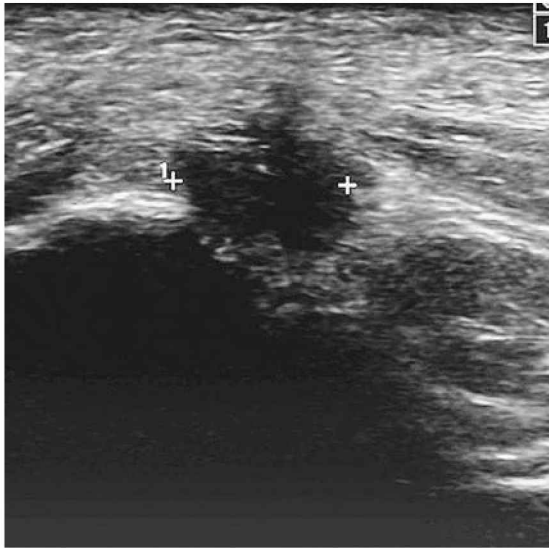
Figure 34.2 (cont.) (e) Palpation. Palpate carefully bearing the underlying anatomy in mind to determine the location of an acute muscle strain (e.g. medial versus lateral hamstring, proximal versus distal etc.) (f) Palpation—gluteal muscles. Palpate the gluteal muscles for trigger points that are taut bands, which are usually exquisitely tender locally and may refer pain into the hamstring muscle (g) Special tests—slump test. The slump test (Chapter 15) is an essential part of the examination of the patient with posterior thigh pain. It helps the clinician differentiate between hamstring muscle injuries and referred pain to the hamstring region from the lumbar spine

7. Special tests

- neuromechanical tests: slump test (Fig.34.2g)
- lumbar spine examination (Chapter 29)
- sacroiliac joint (Chapter 30)
- assessment of lumbopelvic stability (Chapter 11)
- biomechanical analysis (Chapter 8).

Investigations

Investigations of posterior thigh pain can be very useful but clinicians must interpret these findings together with the rest of the examination. Appropriate imaging may include ultrasound (Fig. 34.3a), MRI (Fig. 34.3b) and at times, computed tomography (CT). MRI is the most



(a)



(b)

Figure 34.3 Imaging of hamstring injuries (a) Ultrasound showing hypoechoic area (between electronic callipers, +) (b) MRI demonstrating area of oedema in hamstring consistent with biceps femoris muscle tear

popular option (especially for the elite-level athlete) because it is non-invasive and capable of providing high-resolution images.^{13–16}

PRACTICE PEARL

MRI can help identify the presence and location of pathology, contribute to determining the likely prognosis in certain instances, and monitor healing over time.

Integrating the clinical assessment and investigation to make a diagnosis

Table 34.2 summarises elements of the history, physical examination and investigations that point to whether the diagnosis is likely to be an acute hamstring muscle injury or referred pain to the posterior thigh.

ACUTE HAMSTRING MUSCLE STRAINS

Before outlining type I and type II hamstring injuries, we briefly outline the epidemiology of these injuries

Table 34.2 Clinical features of hamstring muscle tear and referred hamstring pain

Acute hamstring strain (type I or II)	Referred pain to posterior thigh
Sudden onset	May be sudden onset or gradual feeling of tightness
Moderate to severe pain	Usually less severe, may be cramping or 'twinge'
Disabling—difficulty walking, unable to run	Often able to walk/jog pain-free
Markedly reduced stretch	Minimal reduction in stretch
Markedly reduced contraction with pain against resistance	Full or near to full muscle strength against resistance
Local haematoma, bruising	No local signs
Marked focal tenderness	Variable tenderness, usually non-specific
Slump test negative	Slump test frequently positive
May have gluteal trigger points	Gluteal trigger points that reproduce hamstring pain on palpation or needling
May have abnormal lumbar spine/SIJ signs	Frequently have abnormal lumbar spine/SIJ signs

together. As the differentiation of acute hamstring injuries into two types has only recently been recommended, it is not possible to confidently separate epidemiological data.

PRACTICE PEARL

In the large majority of hamstring strains, the injured muscle is biceps femoris (83% of all hamstring injuries). Semimembranosus injuries are uncommon (12%) and semitendinosus injuries are rare (5%).¹

Epidemiology

Acute hamstring strains are common in many popular sports, including the various football codes, field hockey, cricket and athletics. For example, hamstring strains are the most common injury in Australian rules football, constituting 15% of all injuries, with an incidence rate of six injuries per club (approximately 40 players) per season and a cost of 21 missed matches per club per season.¹⁷ Similarly, in English soccer, hamstring strains make up 12% of all injuries, with an incidence rate averaging five injuries per club per season, resulting in 15 matches and 90 days missed.^{18, 19}

Approximately 180 hamstring injuries occur in the 30 teams in the American National Football League (NFL) each year, so the incidence of approximately six injuries per team per season is similar to the AFL. Mean days lost per injury has been found to be 13.2.²⁰ Twelve per cent of all injuries in European elite-level soccer are hamstring injuries. A typical 25-player squad

experiences 4–6 hamstring injuries during a season with a mean of 14.3 days lost per injury (standard deviation 14.9).²¹ A study of American collegiate soccer players noted that men were 64% more likely than women to sustain a hamstring strain.²²

The average injury severity has been reported to be between three to four missed matches per injury.^{17, 23} Hamstring injuries in ballet have not been captured as well as in football codes but estimates of lifetime prevalence of hamstring injury are as high as 51% (34% acute, 17% overuse).²⁴

With respect to epidemiology of recurrence, acute hamstring strains have the highest recurrence rate of any injury, with a rate of 20% in AFL,²⁵ 16.5% in American NFL²⁰ and 16% in European soccer.²¹ It has also been shown in collegiate soccer that recurrence is more common for men than women.²²

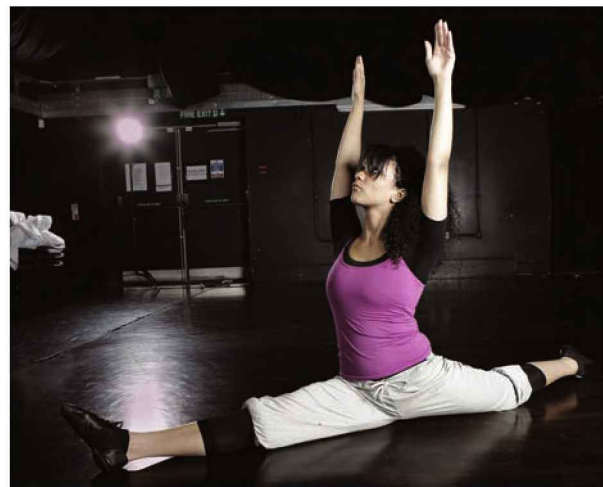
Type I and type II acute hamstring strains— not all acute hamstring injuries are the same!

There are at least two distinctly different types of acute hamstring strains (types I and II), distinguished by different injury situations. The more common, type I, hamstring strains occur during high-speed running (Fig. 34.4a).^{26–32} Type II hamstring strains occur during movements leading to extensive lengthening of the hamstrings in hip flexion combined with knee extension, such as high kicking, sliding tackle and forward split (Fig. 34.4b), and may occur at slow speeds.^{26–29, 31}

Type I strains (the high-speed running type) usually involve the long head of biceps femoris, most commonly at



(a)



(b)

Figure 34.4 (a) Sprinting is the classic activity that causes type I hamstring strains (b) Type II hamstring strains occur with maximal stretching (e.g. dancer's forward split shown with right hamstring under tension)

(A AND B) ISTOCKPHOTO

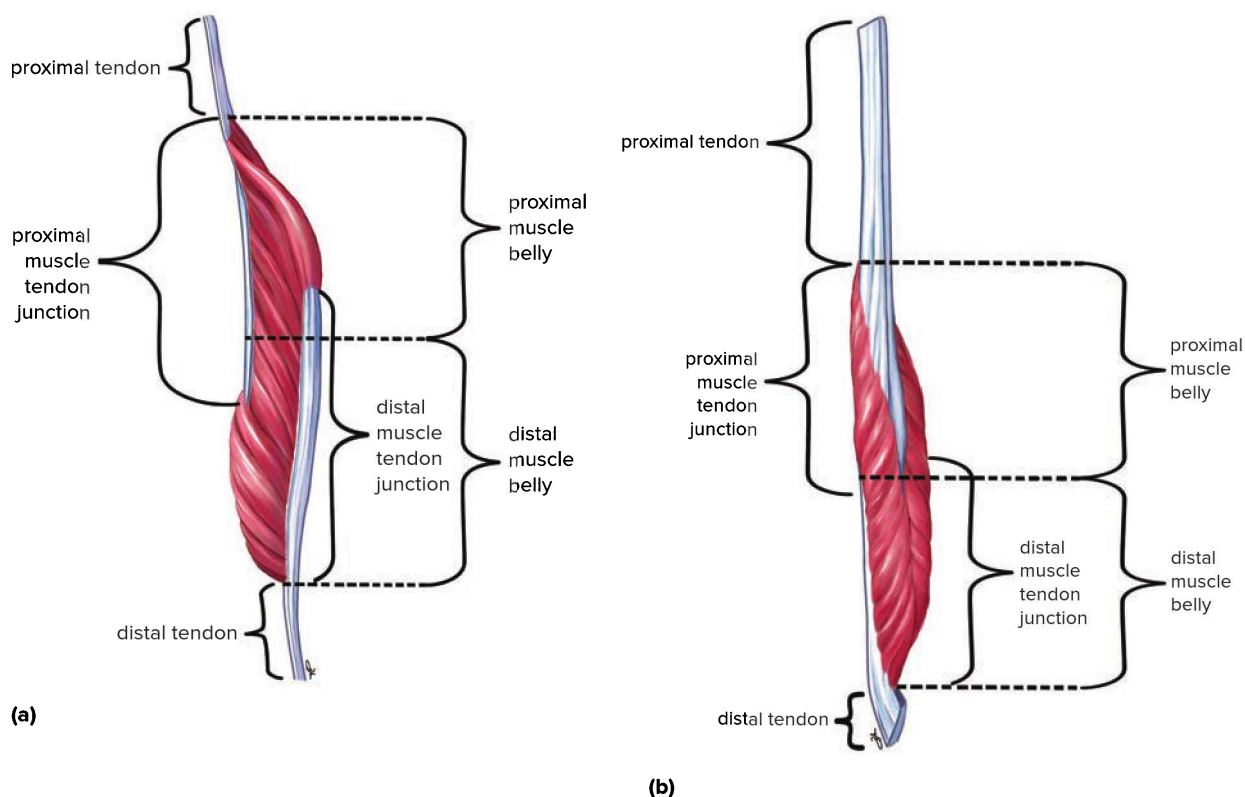
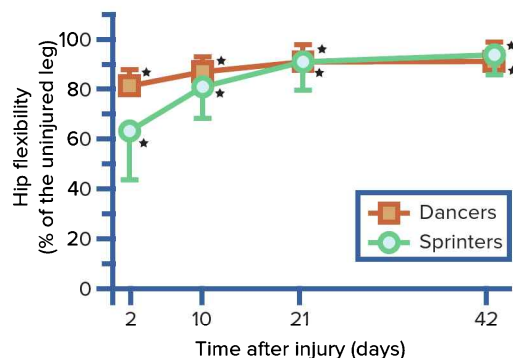


Figure 34.5 (a) Type I strains (the high-speed running type) are mainly located to the long head of biceps femoris and typically involve the proximal muscle–tendon junction (b) Type II injuries (the stretching type) are typically located close to the ischial tuberosity and involve the proximal free tendon of semimembranosus³¹

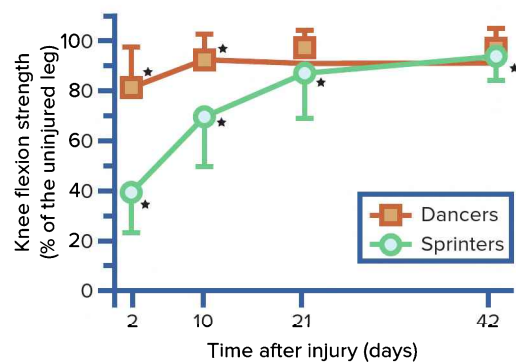
the proximal muscle-tendon junction (Fig. 34.5a).^{26, 27, 30–32} In contrast, type II injuries (the stretching type) are typically located close to the ischial tuberosity and involve the proximal free tendon of semimembranosus (Fig. 34.5b)^{27, 31} thus the stretching type of hamstring strain can in fact be considered a primary tendon injury.³³

Type I strains (e.g. high-speed running) generally cause a more marked acute decline in function, but typically require a shorter rehabilitation period than type II stretching-type hamstring strains (Fig. 34.6).³⁴ The injury mechanism and location can give us important information about the prognosis of the injury.^{26–28, 30, 34} The injury location can be determined both by maximal pain palpation and by MRI during the first 2 weeks after injury occurrence. The closer the site of maximum pain palpation to the ischial tuberosity, the longer the rehabilitation period. An injury site closer to the ischial tuberosity has been reported to be associated with a longer rehabilitation period,^{26, 27, 31, 32, 35} however, a recent study by Moen did not confirm this relationship.¹³ MRI should always be performed when a total rupture is suspected.

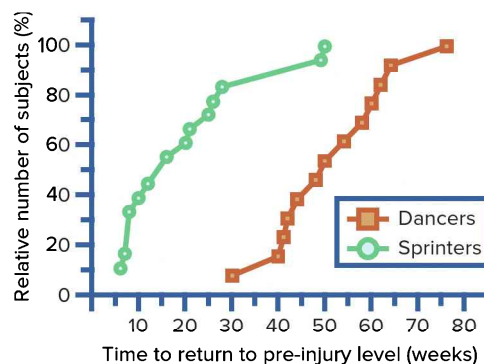


(a)

Figure 34.6 Comparison of hamstring injuries in sprinters (type I) and dancers (type II) (a) Hip flexibility (range of motion) of the injured leg expressed as a percentage of the uninjured leg in the sprinters (n=18) and dancers (n=15). The sprinters' injuries (type I) resulted in more reduction in flexibility, but similar times to return to near pre-injury flexibility levels³⁴ REPRODUCED WITH PERMISSION OF BRITISH JOURNAL OF SPORTS MEDICINE



(b)



(c)

Figure 34.6 (cont.) (b) Knee flexion strength in the injured leg expressed as a percentage of the uninjured leg in the sprinters ($n=18$) and dancers ($n=15$) showing markedly increased reduction in strength in the sprinters' injuries (type I) (c) Relative number of subjects in each group plotted against the corresponding time, in weeks, to return to pre-injury level of performance ($n=18$ for the sprinters type I and $n=13$ for the dancers type II) demonstrating the prolonged rehabilitation time for type II injuries³⁴

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Type I acute hamstring strain: sprinting-related

Although there are a variety of sports skills that can potentially heavily load the hamstrings (e.g. kicking, twisting, jumping, hurdling), sprinting is the most commonly reported mechanism of type I acute hamstring muscle strain.^{8, 23, 36, 37}

Why do hamstrings fail during sprinting? Biomechanical studies suggest that hamstrings are most vulnerable to injury during the terminal swing phase of sprinting,^{38, 39} a time in the stride cycle where they are highly activated as they work eccentrically to decelerate the swinging tibia and control knee extension in preparation for foot strike.⁴⁰

Patients who present with type I acute hamstring strain typically complain of sudden onset pain in the hamstring region that usually stops them. On examination, careful palpation and testing will usually locate the injury in the long head of biceps femoris, often the proximal muscle-tendon junction.^{26, 27, 31, 32} Type I strains are generally associated with a more marked acute decline in function than type II strains.

If imaging is deemed necessary (e.g. if presence of type I acute hamstring strain is unclear on clinical grounds), then MRI (Fig. 34.7) would be the current recommended imaging modality because it is non-invasive and of high resolution. The only disadvantage is cost (rarely an issue for the elite athlete). MRI can be helpful in accurately identifying the location of the injury (semimembranosus versus biceps femoris long head versus semitendinosus;

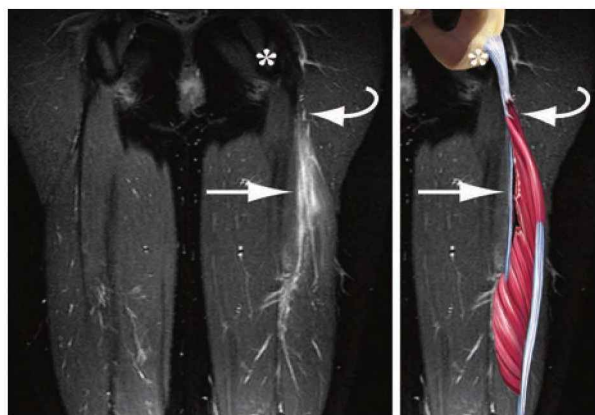


Figure 34.7 MRI showing type I hamstring injury

distal versus proximal; tendon versus muscle-tendon junction versus epimysium).

For two reasons, the role of MRI in predicting prognosis following type I acute hamstring muscle strains is currently uncertain. First, simple clinical parameters (Table 34.2) are just as good if not better predictors of prognosis than MRI-based parameters.^{11, 13} Second, most of the studies investigating the prognostic value of MRI-based parameters are considered to have a high risk of bias and thus findings must be interpreted cautiously,¹⁵ although a recent high-quality study has found increased cranio-caudal length of lesion on MRI to be positively correlated with longer time to RTP.⁴¹ If there is a clinical suspicion of total tendon rupture, then MRI should always be performed.

Type II acute hamstring strain: stretch-related (dancers)

Type II acute hamstring strains more commonly occur in sports that necessitate large amplitude movements and ballistic limb actions, such as ballet dancing and gymnastics. The common mechanism of injury in these instances is an excessive stretch into hip flexion.

In contrast to type I injuries, the stretching-type injuries typically are located close to the ischial tuberosity and involve the proximal free tendon of semimembranosus.^{26, 27, 35, 42} MRI shows the site of the type II injury close to the ischial tuberosity (Fig. 34.8).

Although type II hamstring strains can cause a less dramatic acute limitation than type I strains, their rehabilitation period is often longer than that of type I strains.^{26, 27, 34} It is important to directly inform the athlete that the rehabilitation is likely to be prolonged. Unrealistically optimistic information will only reinforce the disappointment and frustration of the injured athlete. The athlete can often do quite demanding rehabilitation training early on, as long as pain-provoking exercises are avoided. Passive stretching and heavy load exercises seem to aggravate the stretch-type of hamstring injuries by increasing pain.

Prognosis of hamstring injuries

The following factors have been shown to be associated with increased length of time to RTP, although in some instances the evidence is conflicting.

- Time (days) to walk normal pace pain-free has been shown to be significantly associated with RTP time (>1 day = 4 times more likely to take >3 weeks). Combining this with a past history of hamstring injury



Figure 34.8 MRI shows the site of the type II injury close to the ischial tuberosity

within 12 months resulted in a 93% chance of taking longer than 3 weeks to return in elite AFL players.¹² However, note that a more recent study involving athletes from a variety of sports has failed to confirm this relationship.¹³

- Days to jog pain-free has been found to be a strong predictor of time to RTP.⁸
 - 1-2 days $\leq$ 2 weeks to RTP
 - 3-5 days $\geq$ 2 weeks to RTP
 - >5 days $\geq$ 4 weeks to RTP
- MRI-negative 'hamstring strains' are associated with relatively rapid time to RTP and are relatively common (10-20%).^{6, 8, 9, 43}
- Several studies have found the more proximal the site of injury, the more prolonged the time to RTP,^{26, 27, 31} although this relationship was not confirmed in a recent study.¹³
- Injury to the intramuscular ('conjoint' or 'central') tendon of biceps femoris long head is associated with a long time to RTP (e.g. around 3-4 months), much longer than an injury only involving the muscle-tendon junction or epimysium.^{35, 44-46} (See below.)
- A 'kicking' or 'slow-stretching' mechanism of injury is associated with a longer rehabilitation period than a high-speed running mechanism, even though initial signs and symptoms may actually present as far less severe.^{26, 27, 34}
- Greater self-predicted time to RTP.¹³
- Forced to stop training/playing within 5 minutes following onset of pain.¹¹
- Greater maximum pain intensity experienced at the time of injury.^{10, 11}
- Longer length of hamstring tenderness with palpation (cm).¹¹
- Passive range of motion deficit (injured versus uninjured side) recorded within a few days of injury occurrence using the passive straight leg raise test¹³ or the active knee extension test.⁴⁷
- Longer craniocaudal length of lesion on MRI.^{6, 41}

PRACTICE PEARL

Clinicians need to watch out for the type I acute hamstring muscle strain of the proximal biceps femoris long head that is associated with a longitudinal split or partial rupture of the proximal conjoint tendon (as visualised on MRI). These particular injuries require a prolonged rehabilitation period.

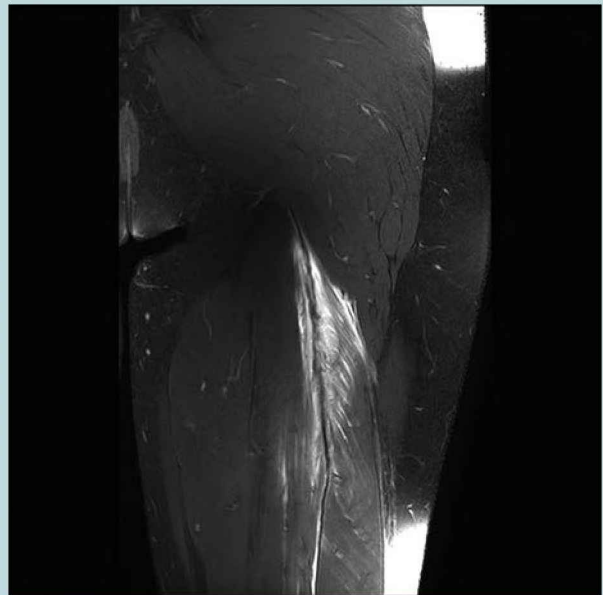
Intramuscular tendon injuries

Injuries to the musculotendinous junction of the proximal or distal intramuscular tendons (Fig. 34.1d) are common. Occasionally the tendon itself is injured. On MRI scanning, the disrupted intramuscular tendon has a wavy retracted appearance as opposed to its normal taut cord-like appearance (Fig. 34.9a). Sometimes the tendon failure occurs like a piece of string, with disrupted tendon strands overlapping each other and offering some scaffold for healing to occur (Fig. 34.9b). In some cases, there is longitudinal delamination and splitting of the intramuscular tendon (Fig. 34.9c), which otherwise appears continuous. In other cases, the tendon may retract and create a gap (Fig. 34.9d).

These injuries have particular clinical significance due to prolonged healing rates and possible increased risk of recurrence.^{44, 46} It is increasingly thought that surgical intervention for this latter pathology offers a predictable and steadfast pathway for healing. An example of a split proximal intramuscular tendon is shown on MRI (Fig. 34.9e) and surgical appearance (Fig. 34.9f). Regardless, of whether conservative or surgical treatment is employed, studies have shown that the RTP time frame is typically 8-10 weeks. High resolution MR scans can help in making the distinction as to whether strands of the disrupted intramuscular tendon rachis remain in broad continuity or are separated and retracted.



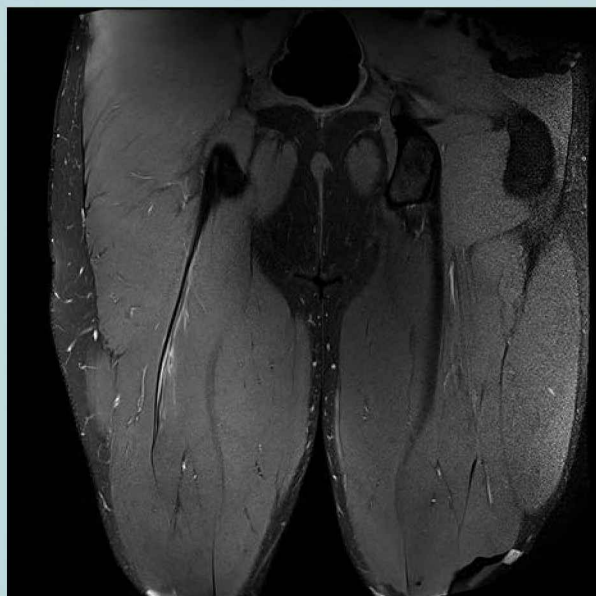
(a)



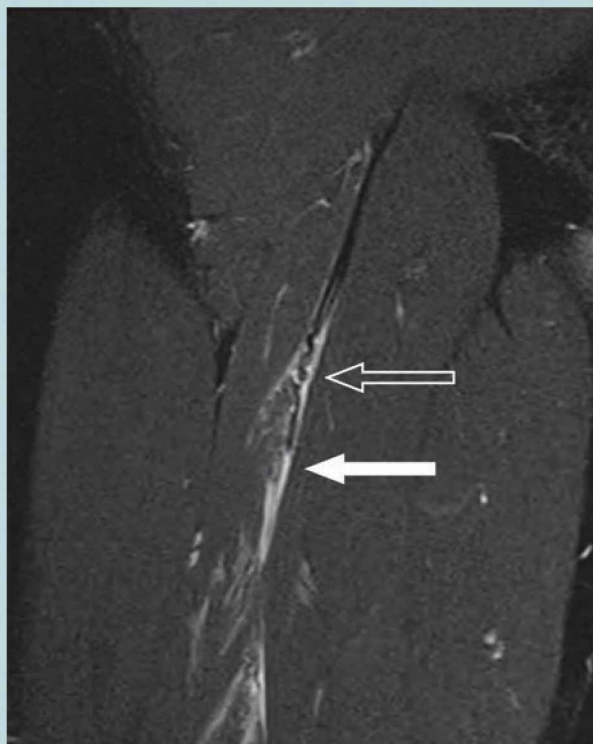
(b)

Figure 34.9 Appearances of injury to the intramuscular biceps femoris tendon (a) A wavy retracted appearance as opposed to its normal taut cord-like appearance (b) Disrupted tendon strands overlapping each other and offering some scaffold for healing to occur

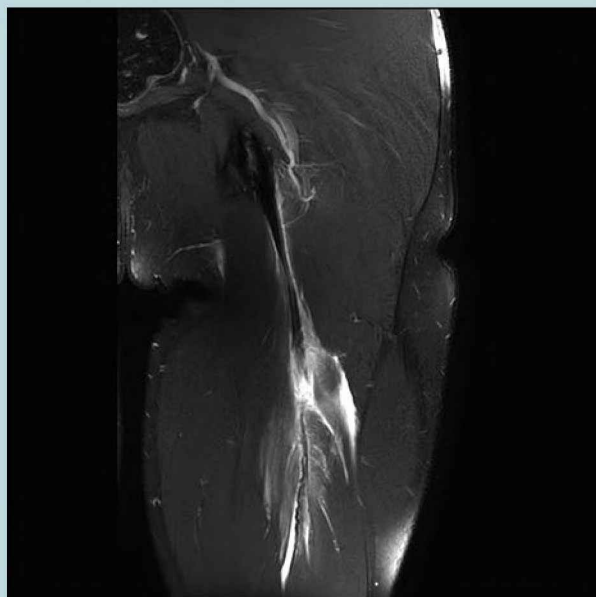
COURTESY OF DR TOM ENTWISLE



(c)



(e)



(d)



(f)

Figure 34.9 (cont.) (c) Longitudinal delamination and splitting of the intramuscular tendon (d) The tendon retracts and creates a gap (e) Disruption of the intramuscular tendon. Compare (e) The MRI appearance of a disrupted intramuscular tendon, showing wavy line indicative of tendon disruption (open arrow) and area of tendon retraction (closed arrow) with (f) The surgical appearance of the same injury. After surgery to repair the defect, the athlete made a successful RTP at 9 weeks (a-d) COURTESY OF DR TOM ENTWISLE (f) COURTESY OF JULIAN FELLER

Management of hamstring injuries

Six RCTs evaluated the efficacy of a particular intervention for the rehabilitation of acute hamstring

strains.^{26, 27, 41, 48–50} While these studies have provided clinicians with some important information, more research is required to fully elucidate the best treatment

PRACTICE PEARL

Progression through phases of rehab program must NOT be time-dependent. Sometimes what initially appears to be a rather minor injury can take an extended period to fully recover and vice versa. Progression must be based on successfully achieving key functional and/or clinical criteria (see Table 34.3).

options, particularly with regards to avoiding recurrence following RTP. Hence, the approach below remains largely based on clinical experience. Management of acute hamstring strains is summarised in Table 34.3.

Rehabilitation programs require a basic structure, but should never be a 'recipe'. Treat each case on its merits. The following is a suggested structure, which is similar to that recommended by other experienced clinicians.^{43, 51}

Acute management phase

Acute injuries should always be assessed thoroughly before any treatment is administered. The fundamental objective of the acute management phase is to facilitate myofibre regeneration and minimise fibrosis. If strategies aimed at minimising scar tissue formation are instituted immediately, then such strategies may reduce the chances of injury recurrence.

Rest, ice, compression, elevation

Traditionally, the most common treatment in the first few days or thereabouts following type I acute hamstring strain is the rest, ice, compression, elevation (RICE) program. For example, applying ice for 10–15 minutes using a cold pack, every 3–4 hours, for the first few days until acute symptoms settle. Compression can be achieved in-between times via an elastic bandage or Tubigrip™ stocking.

Muscle activation

Although RICE would still be the recommended initial approach, recent research in cell therapy and tissue engineering is indicating an additional role for controlled and monitored exercise (or muscle contraction) regimens. Muscle contraction promotes angiogenesis (i.e. the formation of new blood vessels and the expansion of existing vascular trees) and in doing so increases the likelihood of delivering muscle-derived stem cells to the injured region. These cells are likely derived from the vascular endothelium and offer great potential for providing long-term myofibre regeneration.⁵² Another aim of early pain-free muscle contractions is to prevent neuromuscular inhibition around the injured area. Note that this recommended

approach to muscle healing is consistent with the concept of 'mechanotherapy'.⁵³

The commencement of frequent (e.g. 3–4 times per day), low-grade, pain-free muscle contractions (e.g. simple isometric hamstring contractions or active prone knee bends) immediately following injury would appear advantageous (Fig. 34.10). Such exercises should be biased as much as possible towards selective activation of the injured muscle.

Medical therapies

A number of different medical therapies have been used in the management of acute hamstring muscle injuries. There is little evidence to support their use.⁵⁴

Despite the widespread use of nonsteroidal anti-inflammatory drugs (NSAIDs) in the treatment of hamstring injuries, the two randomised controlled studies



(a)



(b)

Figure 34.10 Early pain-free muscle contraction (a) Single-leg hip extension exercise with whole leg a few centimetres off the plinth (b) Active prone knee bends

failed to show beneficial or superior effects of NSAIDs compared to analgesics or placebo on acute muscle strain injuries.^{12, 55} It is likely that simple analgesics are just as effective without the long-term risks on skeletal muscle and the gastrointestinal system associated with NSAIDs.

One study showed favourable results with intramuscular corticosteroid injection in American football players with acute hamstring injuries.⁵⁶ Previously, the use of corticosteroids in acute muscle strains had been clearly contraindicated because they were thought to delay elimination of haematoma and necrotic tissue, as well as retard muscle regeneration. There are concerns regarding the retrospective nature of the NFL study and lack of control group, so we caution against the use of corticosteroids in this situation in light of the fact that there have been no further studies confirming these results.

The pros and cons of delivering growth factors via autologous blood or platelet-rich plasma (PRP) injections are discussed in Chapter 17. A number of studies have investigated the effect of PRP administration specifically in hamstring muscle injuries.

One study suggested that a single autologous PRP injection combined with a rehabilitation program was significantly more effective in treating hamstring injuries than a rehabilitation program alone;⁵⁷ however, a number of subsequent studies of higher quality have shown no significant benefit.^{58–60} At this stage, there is insufficient evidence to recommend the use of PRP in acute hamstring muscle injuries.

The combined injection of Traumeel[®] S and Actovegin[®], a deproteinised calf-blood haemodialysate, immediately after a hamstring muscle injury and again at days 2 and 4 post-injury to the area of the muscle strain and the lumbar spine is common practice in sports medicine in Germany, and has been used elsewhere in some instances, despite the lack of any controlled trials supporting its use.^{61, 62} A pilot study with limited subjects and a poor methodology reported a significant difference in subjective recovery and the RTP time in athletes following the intramuscular administration of Actovegin into grade I muscle tears, but there is no further evidence available.⁶³

In summary, there is currently no good scientific evidence to support the use of any medication in the management of hamstring muscle injuries.

Criteria for progression to subacute phase

Once the athlete is pain-free with walking and adequate/sufficient force can be generated without pain on resisted hamstrings contraction, then jogging can commence and gradually increased.

In some instances, the athlete can experience considerable improvement in clinical signs and symptoms within 4–6 days post-injury. However, the clinician must remember that the healing process is still in its initial stages and the risk for re-injury is high, as the injured tissue is unlikely to be able to tolerate maximal loads. The criteria for progression to the subacute phase are shown in Table 34.3.

Table 34.3 Criterion-related rehabilitation of hamstring muscle strains

Phase	Key criteria
1. Begin running/active rehabilitation (i.e. begin subacute phase)	Pain-free walking Adequate force with resisted muscle contraction
2. Return to full training (i.e. begin functional phase)	Complete resolution of any symptoms with maximal resisted muscle contraction Equivalent tenderness upon palpation (left = right) Full and symptom free range of movement/flexibility (left = right) Successful completion of a structured running program (i.e. maximum sprinting speed achieved at the end of the running program is comparable to the athlete's previously recorded benchmark when uninjured) Successful completion of appropriate rehabilitation exercises Successful completion of controlled functional (sports-specific) tasks simulating the original injury mechanism
3. Return to play (RTP)	Successful completion of at least one full week of normal training load with no adverse reaction in any clinical and/or functional signs and symptoms Additional quantitative tests of hamstring function (e.g. hand-held dynamometry, isokinetic dynamometry, Askling H-test (below), the single-leg bridge capacity test, performance on Nordic eccentric hamstring exercise); compare data for injured limb to uninjured limb or to data collected prior to injury No adverse psychological characteristics, such as apprehension regarding RTP or fear of re-injury

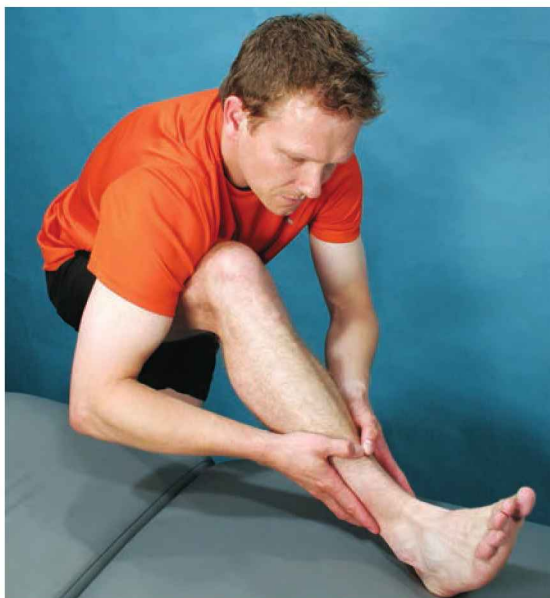
Subacute/conditioning phase

Stretching

The role of targeted hamstring stretching in the rehabilitation of hamstring strain injury should be considered on an individual basis (Fig. 34.11). There is some limited evidence to suggest that rate of recovery can be increased with an increased daily frequency of hamstring-stretching exercises.⁴⁹ Consequently, to prevent long-term loss of range of motion (e.g. perhaps from significant scar tissue), a controlled stretching program can be instituted. However,



(a)



(b)

Figure 34.11 Hamstring stretches (a) Hamstring stretch with contralateral knee flexion. The lower leg can be placed in different degrees of external and internal rotation to maximise the effectiveness of the stretch (b) Hamstring stretch with bent knee results in maximal stretch to the upper hamstrings

in our clinical experience, most athletes regain their normal range of motion without the need for excessive or aggressive hamstring-stretching regimens.

It may be more important to focus on stretching of other structures. For example, tight hip flexor muscles, if present, may place the athlete at greater risk of hamstring strain.^{64, 65}

Manual therapy

A comprehensive clinical examination of the lumbar spine, sacroiliac and buttock regions should be instituted at an early stage to assess whether or not these regions have any contribution to the presenting injury. For example, Cibulka et al. reported that mobilisation of the sacroiliac joint was of some benefit in the treatment of acute hamstring strains.⁴⁸

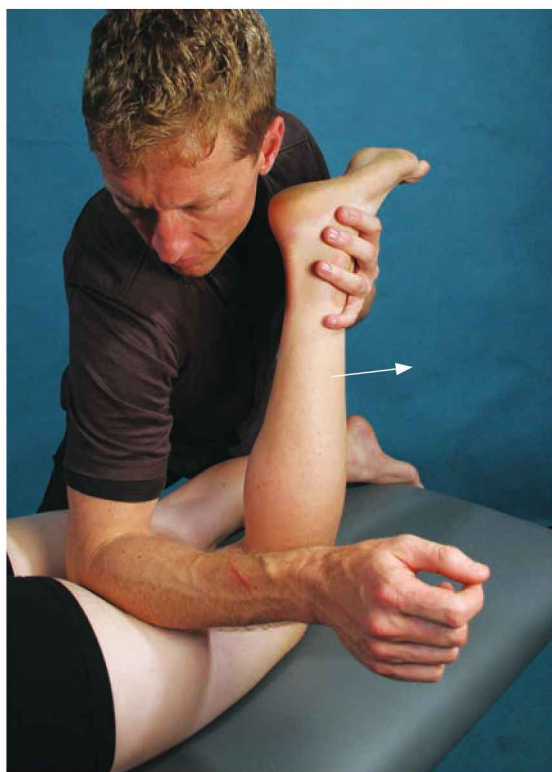
Neural mobility restriction is frequently present in hamstring injuries secondary to bleeding around the sciatic nerve. Neural mobilising exercises should be performed to reduce adhesions. Neural mobilising can be performed in the hamstring stretch position (Fig. 34.11) by adding gentle cervical flexion.

Soft tissue techniques can be used in the treatment of hamstring strains. Sustained compression (Fig. 34.12a) and sustained myofascial tension (Fig. 34.12b) are used, gently at



(a)

Figure 34.12 Soft tissue techniques in the treatment of hamstring injuries (a) Sustained compression force to hamstring



(b)

Figure 34.12 (cont.) (b) Sustained myofascial tension combined with passive knee extension. The hand or the elbow (illustrated) is kept stationary and the release is performed by passively extending the knee (arrow)

first and then more vigorously. Longitudinal massage along the muscle may assist in optimising scar reorganisation. Abnormalities of the gluteal muscles may be associated with hamstring strains. These regions may be treated in a side-lying position using elbow ischaemic pressure with the tissue on stretch (Fig. 34.13).

Strengthening for hamstring muscles

Strengthening is an essential component of the rehabilitation and prevention of hamstring injuries. When designing a strengthening program, the clinician needs to consider the following parameters: the type of contraction (i.e. eccentric/isometric/concentric), the length of the muscle-tendon unit (i.e. lengthened/optimal/shortened) and the speed of contraction (i.e. slow/fast rate of force development). A further consideration is selective recruitment (i.e. medial versus lateral), as different exercises tend to preferentially activate different parts of the hamstring muscle group.^{66–68} A comprehensive strengthening program will therefore require a variety of exercises.



Figure 34.13 Treatment of the gluteal region in a side-lying position using elbow ischaemic pressure

Most hamstring strain researchers probably agree that the strains occur when the hamstring muscle group is extensively lengthened, especially in the stretching-type^{35, 42} injury, but also in the sprinting type of injury.^{40, 69}

It is also generally accepted that retraining needs to be specific to muscle function and as much as possible to the specific hamstring muscle injured (i.e. most commonly biceps femoris long head).^{70–73}

Based on this concept, training programs for the rehabilitation of hamstring injuries should include exercises emphasising eccentric muscle contractions and extensive lengthening. It is also recommended to use exercises that load the hamstrings about the hip and the knee simultaneously, since three of the four hamstring muscles are bi-articular.

Recent clinical trials have shown that a rehabilitation program emphasising lengthening-type exercises (L-protocol) is more effective after acute hamstring injuries in elite footballers/sprinters/jumpers than a program

B

containing inner range exercises such as supine single leg bridging and standing hip extension against a pulley.^{26, 27}

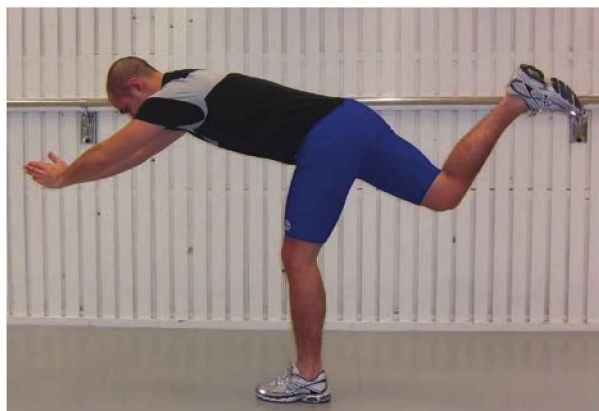
The L-protocol (lengthening) consists of three exercises (the extender, the diver and the glider) specifically aimed at loading the hamstrings during extensive lengthening, shown and explained in detail in Figures 34.14. The extender is aimed mainly at increasing flexibility, the diver is a combined exercise for hamstring strength and trunk/pelvis stabilisation, and the glider is more of a specific eccentric hamstring exercise. The L-protocol could be started 3–5 days after injury occurrence and no pain should be provoked.^{26, 27}

Another recent clinical trial⁴¹ compared a rehabilitation program containing progressive running and eccentric strengthening to a program containing progressive agility and trunk stabilisation exercises. The time taken to RTP was similar for both programs.

There are a wide variety of strengthening exercises that load the hamstrings eccentrically and involve simultaneous hip and knee joint motions. Figure 34.15 illustrates some of the more popular exercises:



(a)



(b)



(c)

Figure 34.14 The L-lengthening protocol recommended by Askling (a) The Extender (aimed mainly at increasing flexibility) The subject should hold and stabilise the thigh of the injured leg with the hip flexed approximately 90° and then perform slow knee extensions to a point just before pain is felt. Perform twice every day, 3 sets with 12 repetitions (b) The Diver (combined exercise for hamstring strength and trunk/pelvis stabilisation) The exercise should be performed as a simulated dive, that is, as a hip flexion (from an upright trunk position) of the injured, standing leg and simultaneous stretching of the arms forwards and attempting maximal hip extension of the lifted leg while keeping the pelvis horizontal; angles at the knee should be maintained at 10–20° in the standing leg and at 90° in the lifted leg. Due to its complexity, this exercise should be performed very slowly in the beginning. Perform once every other day, 3 sets with 6 repetitions (c) The Glider (a specific eccentric strength exercise). The exercise is started from a position with upright trunk, one hand holding onto a support and legs slightly split. All the body weight should be on the heel of the injured (here left) leg with approximately 10–20° flexion in the knee. The motion is started by gliding backwards on the other leg (note low-friction sock) and stopped before pain is reached. The movement back to the starting position should be performed with the help of both arms, not using the injured leg. Progression is achieved by increasing the gliding distance and performing the exercise faster. Perform once every third day, 3 sets with 4 repetitions

- REPRODUCED WITH PERMISSION OF BRITISH JOURNAL OF SPORTS MEDICINE
- standing single leg hamstring catches with TheraBand™ (Fig. 34.15a)
 - single-leg bridge catch (Fig. 34.15b)
 - single-leg ball roll-outs (Fig. 34.15c)
 - single-leg slide-outs (Fig. 34.15d)
 - Nordic drops (Fig. 34.15e)
 - single-leg reverse deadlift with kettle bell (Fig. 34.15f)
 - Yo-Yo (Fig. 34.15g)
 - L-protocol: extender, glider, diver (Fig. 34.14).



(a)



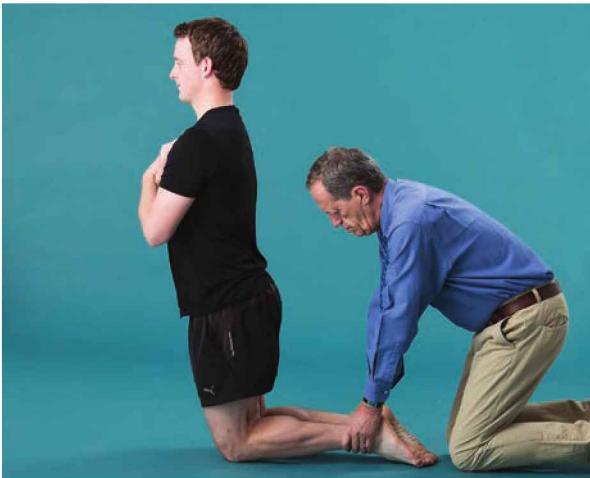
(b)



(c)



(d)

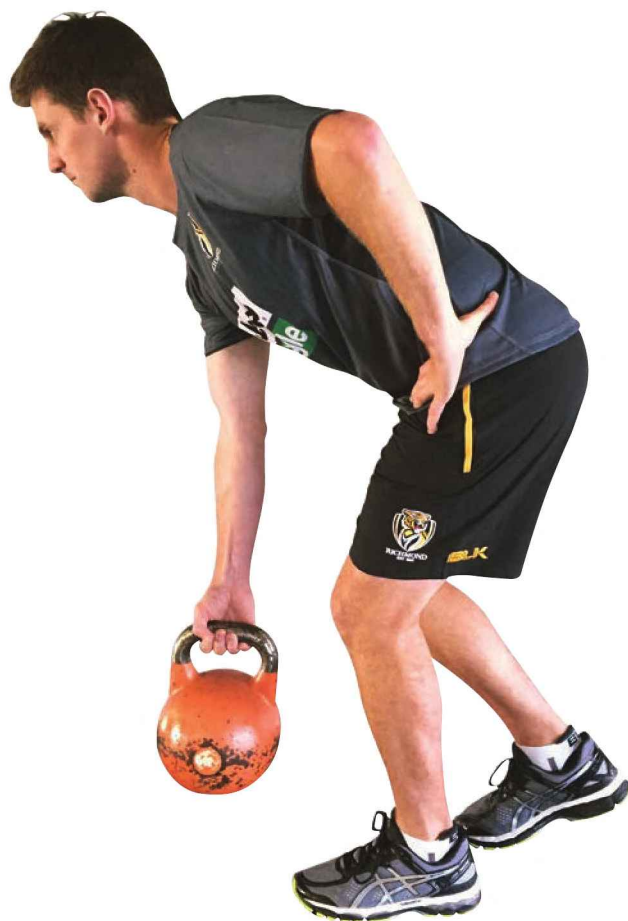


(e-i)



(e-ii)

Figure 34.15 Strengthening exercises (a) Standing single-leg hamstring catches with TheraBand™ (b) Single-leg bridge catch (c) Single-leg ball roll-outs (d) Single-leg slide-outs (e) Nordic eccentric hamstring exercise (drops): patients allow themselves to fall forwards and then resist the fall for as long as possible using their hamstrings



(f)

One study has confirmed the efficacy of the Nordic drops in developing hamstring strength.⁷⁴ Other studies have shown that Nordic exercises are effective in preventing both new and recurrent hamstring injuries (see below).^{75–77}

The Nordic strengthening program shown in Table 34.4 is based on the Mjolsnes and Arnason studies.^{74, 75} It is designed for a 5–10 week pre-season training program. Because of the eccentric load, it is important to ensure that there is at least one day between sessions and that the loads are adjusted if excessive delayed onset muscle soreness (DOMS) results, as this may affect compliance. If the program is continued in-season, then one session per week is adequate.

Generally, low grade/minor hamstring injuries or first-time injuries will progress quickly; therefore, functional strength may be adequate to allow return to sport. In this situation, an exercise such as the single-leg bridge catch (Fig. 34.15b) is probably sufficient.



(g)

Figure 34.15 (cont.) (f) The single-leg reverse dead lift.

The athlete holds a kettle bell in one hand and flexes forward at the hip while maintaining the lumbar spine in a neutral position. Note that in this instance the non weight-bearing limb is NOT used as a counter lever to specifically increase the load on the hamstrings of the weight-bearing limb. The exercise can be progressed by increasing the weight of the kettle bell, the depth of movement (i.e. the amount of hip flexion) and the speed of movement (i.e. drop/catch) (g) Yo-Yo machine

More severe or recurrent injuries require more advanced strength exercises and high-level eccentric loading (e.g. Nordic eccentric hamstring exercise, single-leg reverse dead lift, L-protocol). Eccentric muscle training results in muscle damage and DOMS in those unaccustomed to it. Therefore, any eccentric strengthening program should allow adequate time for recovery, especially in the first few weeks.

Strengthening for hamstring synergists

Rehabilitation must not be restricted to the hamstring muscles alone—it must also include exercises to target muscles that assist the hip extensor function of the hamstrings, such as the gluteus maximus and adductor magnus muscles. Thus, if gluteal and adductor magnus strength is inadequate in a sprinting athlete, it is possible that the hamstrings will be overloaded.

The gluteus maximus acts during running to control trunk flexion of the stance leg, decelerate the swing

Table 34.4 Pre-season training protocol for Nordic hamstring exercises

Week	Sessions per week	Sets and repetitions	Load
1	1	2 × 5	Load is increased as subject can withstand the forward fall longer. When managing to withstand the whole range of motion for 12 repetitions, increase load by adding speed to the starting phase of the motion. The partner can also increase loading further by pushing at the back of the shoulders
2	2	2 × 6	
3	3	3 × 6–8	
4	3	3 × 8–10	
5–10	3	3 × 12/10/8	

leg and extend the hip.⁷⁸ Therefore, any alteration in gluteus maximus activation, strength or endurance may place greater demand on the hamstrings to control hip extension of the stance leg and decelerate the leg during the swing phase. Overall, the gluteus maximus provides powerful hip extension when sprinting.⁷⁹

The hip extensor moment arm of the gluteus maximus is largest when the hip is in extended, whereas the hip extensor moment arm of the adductor magnus is largest when the hip is flexed.⁸⁰ Thus, the gluteus maximus is a strong hip extensor of the extended hip, whereas the adductor magnus is a strong hip extensor of the flexed hip.

To improve gluteus maximus activation, strength and endurance, initially teach good motor patterns and isolate the gluteus maximus from hamstrings, commencing bridges with both legs then progressing to one leg, and finally progress to high load hip extension exercises such as the barbell hip thrust (Fig. 34.16a).⁸¹ Other hip extensor strength exercises that load both the gluteus maximus and adductor magnus in addition to the proximal hamstrings include squats (Fig. 34.16c) and deep lunges (Fig. 34.16d).^{66, 79, 82}

Neuromuscular control exercises

Neuromuscular control of the lumbopelvic region, including anterior and posterior pelvic tilt, may be required to promote optimal function of the hamstrings in sprinting and high-speed skilled movement. For example, a rehabilitation program focusing on progressive agility, neuromuscular control and lumbopelvic stability exercises has been shown to be more effective in preventing injury recurrence than a more traditional stretching and simple strengthening exercise program.⁵⁰ Furthermore, the time taken to RTP appears to be similar for progressive agility and trunk stabilisation exercises compared to progressive running

and eccentric strengthening.⁴¹ Progressive agility and trunk stabilisation exercises include fast feet drills, side-stepping, karaoke side-stepping, single-leg balance



(a)



(b)

Figure 34.16 Gluteus maximus and adductor magnus strengthening exercises: barbell hip thrust (a) starting with hips flexed and (b) moving hips into extension



(c)



(d)

Figure 34.16 (cont.) (c) squat (d) deep lunge

(Fig. 34.17), single-leg stand windmill touches (moving in and out of a body weight arabesque position) and supine single-leg bridging.^{41, 50}

It may also be important to consider motor control exercises for the deep lumbar spine stabilisers (e.g. multifidus). Such exercises can involve isolated activations in prone lying, and then progress to functional postural control exercises against gravity (e.g. sit-to-stand action, inclining the trunk forwards by flexing at the hip while maintaining an extended posture at the lower lumbar spine, performed against resistance). Recent studies have found motor control training to be effective in reducing lower-limb injuries in Australian rules football players.^{83, 84}

Other examples of advanced neuromuscular control exercises for the lumbopelvic region and entire lower extremity are discussed in Chapter 11.

Functional progression

Early commencement of a structured running program is an important part of the rehabilitation

**Figure 34.17** Lumbopelvic stability exercises—single-leg balance. A quarter squat can be added

Key principles to consider when designing a running program following hamstring muscle strains

1. Running sessions may be completed 5–6 times per week in the initial stages of the rehabilitation process when volume and intensity is low (if so desired). However, once the sessions become more advanced, a day on/day off approach is recommended (i.e. 3 times per week). Running on a day on/day off basis allows the athlete sufficient time to recover and adapt between sessions. The clinician can assess key clinical signs the following day and make a judgment as to whether or not the athlete has tolerated or reacted adversely to the load. The structure of the next running session can then be planned accordingly.
2. Running can commence once the athlete is capable of producing adequate force with resisted muscle contraction and has no pain with walking or other everyday activities. Begin with slow jogging intervals and then progress to fartlek-type running as tolerated (e.g. 15–20 repetitions of a 10-second walk followed by a 20-second jog followed by a 30-second run).
3. Once the athlete is capable of achieving approximately 50% pace without symptoms or apprehension, then running may be progressed to a structured running program that involves repeated strides over a distance of 60–90 m at a prescribed intensity. In order to increase the dose of high-speed running in the latter stages of the rehabilitation process, extra interval running at a safe intensity can be completed in addition to the structured running program (e.g. 150 m or 200 m repetitions).
4. Load on the hamstrings increases with faster running.⁴⁰ Therefore, running speed both within a session and from one session to the next should always be progressed in a graduated manner. Rapid increments in running intensity must be avoided, especially during the early stages of the rehabilitation process.
5. The warm-up prior to running should consist of jogging as well as footwork and agility drills.^{41, 50}
6. Sprinting technique drills should also be integrated into the running program (e.g. as part of the warm-up). Sprinting technique drills have been shown to be effective in improving an athlete's lower-limb joint position sense (or proprioception).⁸⁵ Furthermore, poor lower-limb joint position sense has been linked with greater risk of hamstring strains in Australian rules football players.⁸⁶
7. Sport-specific training drills should be included in the final stages of the rehabilitation process.⁸⁷ Ideally, these drills should simulate as closely as possible the original mechanism of injury.

process following a hamstring muscle strain. The basic principles of the running program are outlined in the box above. Once the athlete has successfully completed all running progressions as well as relevant sport-specific activities, and all strength and neuromuscular control exercises have been advanced to a sufficient level, then the athlete can progress to the next stage—the RTP phase.

PRACTICE PEARL

Progression of hamstring strain rehabilitation is NOT time dependent, but rather progressed on the basis of clinical signs (key outcome measures).

RTP phase

It is extremely difficult to decide when the athlete is ready to RTP after a hamstring strain.^{2, 43, 50, 88} These difficulties may be reflected in a conspicuously high injury recurrence rate, particularly within a few weeks after the return.^{36, 89} This vulnerability to strain persists, though gradually reduces, for many weeks following RTP.¹⁷

RTP rehabilitation programs that rely solely on subjective measures such as 'pain-free movements' may result in deficits in neuromuscular control, strength, flexibility, ground reaction force attenuation and production, and lead to asymmetries between legs during normal athletic movements.⁷⁹ These deficits and deficiencies could persist into sport practice and competition, and ultimately increase the risk of re-injury and limit athletic performance.

A criteria-based approach to rehabilitation that includes objective and quantitative tests has the potential to identify deficits and address them in a systematic progression (i.e. algorithm) during the stages of returning to sport. While validated criteria to assess whether an athlete can safely RTP are still lacking, criteria that should be considered by the clinician when making this assessment include:

- no difference in tenderness with palpation for the injured versus uninjured hamstring¹⁴
- full range of movement (equal to uninjured leg) measured via:
 - active/passive straight leg raise test^{90, 91}
 - active knee extension test¹⁴
- pain-free maximal isometric contraction

- objective strength assessments measured via:
 - hand-held dynamometer⁹² or isokinetic dynamometer^{93, 94}
 - performance with the Nordic eccentric hamstring exercise (e.g. using the NordBord⁸⁸ to measure the force from each lower limb independently during the exercise—see below)
 - performance with the single-leg bridge capacity test⁹⁵
- Asking's H-test⁹⁶ (see box below)
- completion of structured running program:
 - maximum sprinting speed achieved at the end of the running program is comparable to the athlete's previously recorded benchmark when uninjured
- functional tests:
 - repeat maximum accelerations from a standing start over a variety of distances (e.g. 5 m, 10 m, 15 m, 20 m)
 - high-intensity weaving/cutting and random (unanticipated) agility activities
 - successful execution of sport-specific activities (e.g. bending over to gather a ground ball while running at speed for Australian rules football players)
- successful completion of a sufficient amount of standard training sessions (e.g. 1 to 2 weeks)
- evaluation of an athlete's psychological status regarding readiness for RTP and fear of re-injury.

While it is recommended that athletes do not RTP following hamstring strains until all clinical signs and

Asking's H-test

Asking's H-test is a complement to the clinical examination before return to sport.⁹⁶ Notably, this active test must not be performed before all clinical tests, including those of passive flexibility, indicate complete recovery. During the test, the subject should be positioned on a bench in a supine position with the contralateral leg and the upper body stabilised with straps (Fig. 34.18). A knee brace ensures full knee extension of the tested leg and the foot of the tested leg should be kept in a slightly plantar-flexed position. No warm-up exercises are to be performed before the test. The uninjured leg is tested before the injured leg.

The instruction to the subject is to perform a straight leg raise as fast as possible to the highest point without taking any risk of injury. A set of three consecutive trials are performed, preceded by one practice trial with submaximal effort. After the three active test trials, the subject is to estimate his or her experience of insecurity and pain on a VAS scale, from 0 to 100. In the study by Asking, Nilsson and Thorstensson, the athletes noted an average insecurity estimation of 52 for the injured leg and 0 for the uninjured leg, respectively.⁹⁶

The active test seems to be sensitive enough to detect differences both in active flexibility and in insecurity after acute hamstring strains at a point in time when the commonly used clinical examination fails to reveal injury signs. If insecurity persists, the test should be repeated until no insecurity is reported. The athlete is then allowed to return to sport.



(a)



(b)

Figure 34.18 Asking's hamstring apprehension test
(a) Starting position (b) Maximal hip flexion

symptoms have fully resolved,¹⁴ clinicians should recognise that many athletes are capable of coping with advanced functional progressions and making a successful RTP despite in some instances having residual strength deficits of 10% of more^{94, 97} and persisting increased signal intensity on MRI.^{6, 97, 98} These characteristics may take up to 6 months to fully resolve.⁹⁷ Hence, MRI appearance does not seem to be a valid indicator of readiness to RTP.

MRI can monitor tissue healing, but the extent of healing on MRI that is sufficient for an athlete to RTP with minimal risk for recurrence is presently not known. It is therefore evident that RTP following hamstring muscle strains rarely coincides with complete healing from the injury, making the timing of RTP a very challenging management decision for the clinician. The issue of RTP has been suggested to be one of risk minimisation (not risk elimination), as there appears to be trade-off between quick RTP and likelihood for recurrence.⁹⁹ This trade-off must be acknowledged and carefully considered by the clinician on a case-by-case basis.

The length of time until RTP depends on the severity of the injury. In most cases, an athlete with a mild hamstring strain who is optimally managed will be able to successfully RTP within approximately 18 days.¹

A practical tip to reduce the incidence of recurrence is to restrict game time when first returning from hamstring injury to minimise fatigue and reduce overall exposure to high-risk activities. In this way, RTP can be gradually progressed over several weeks.

PRACTICE PEARL

It is important to persist with a well-structured strength and neuromuscular control exercise program after RTP to lessen the likelihood of recurrence. All elements of the program should be continued until all impairments have been resolved. These strength sessions must be carefully scheduled to allow recovery time before exposing the athlete to high-risk activities again.

RISK FACTORS FOR ACUTE HAMSTRING STRAIN

We discuss risk factors here because patients rarely present for 'primary prevention'—to avoid hamstring injuries before they have one. We review risk factors for acute hamstring strain to identify those which may be mitigated. As with all injuries (Chapters 3 and 4),

risk factors for acute hamstring strains may be intrinsic (person-related) or extrinsic (environment-related) factors. Two systematic reviews were published on this topic in 2013.^{100, 101}

Factors that appear to predict risk of hamstring strain are discussed in the following section. Other factors have been evaluated, but do not appear to be risk factors. Such factors include:

- body mass index^{36, 102, 103}
- height^{36, 37, 102, 103, 105}
- weight^{36, 102, 103, 105}
- functional performance tests (e.g. countermovement jump, 40 m sprint tests, gross performance with the Nordic eccentric hamstring exercise, hamstring length measurement).¹⁰³

Intrinsic risk factors

Age

A number of studies have shown that athletes of an older age are at increased risk of acute hamstring strain, even when the confounding factor of previous injury is removed.^{37, 102–105} Australian rules football players aged 23 years or more were almost four times as likely to sustain an acute hamstring strain before their younger counterparts during the season.³⁷

Older athletes may be at heightened risk due to increased body weight and reduced hip flexor flexibility.⁶⁵ It has also been speculated that the high risk of older athletes to acute hamstring strain is related to degenerative changes at the lumbosacral junction.¹⁰⁶

Past history of injury—a critical factor

Past history of injury is a critical factor for the development of another hamstring injury. It has been well demonstrated that a prior history of acute hamstring strain is a strong risk factor for future injury.^{100, 101} In comparison to athletes without a history of hamstring strain, recent studies have found that those with a history do not appear to display appreciable differences in running mechanics;^{107–109} however, they do display evidence of neuromuscular inhibition,^{110–112} specific strength deficits,^{110, 112} as well as altered muscle-tendon morphology (muscle atrophy, reduced muscle fascicle length, presence of scar tissue)^{97, 113, 114} and contraction mechanics.¹¹⁵

It may be that future risk of re-injury in athletes who have suffered a hamstring strain is explained by the presence of one or more of these characteristics. Clinical interventions (e.g. exercise, manual therapy) aimed at addressing or minimising the impact of such characteristics are therefore of great importance.

Athletes with a history of injury to other areas of the lower limb also have an increased likelihood of acute hamstring strain. These include:

- knee—major knee injury (anterior cruciate reconstruction, patellar dislocation)^{105, 116}
- groin—history of groin pain (bone oedema MRI)¹⁰⁵
- calf muscle strain¹⁰⁴
- lumbar spine—major injury (i.e. episode which required radiological investigation with a specific recorded clinical diagnosis).¹⁰⁵

Strength

Numerous studies have evaluated whether strength quantified via an isokinetic dynamometer is a risk factor for hamstring strain. While intuitively reduced muscle strength would appear to predispose to hamstring strain, the evidence is mixed. Such conflicting findings may relate to the difficulty in quantifying muscle strength in a systematic manner; that is, what type of contraction (isometric, concentric, eccentric), what strength parameter or index to measure (peak torque, knee angle at which peak torque occurs, hamstring/quadriceps ratio, hamstring to hip flexor ratio, strength asymmetry), as well as the level of motivation of the athletes.

A recent systematic review and meta-analysis evaluating the existing evidence regarding risk factors for hamstring strain reported a somewhat counterintuitive outcome concerning strength; specifically, that increased quadriceps strength is a risk factor for hamstring strain but reduced hamstring strength is not.¹⁰⁰ This result might be a consequence of some of the previously outlined issues with measuring strength, but it might also be a consequence of previous risk factor studies not taking the multifactorial nature of hamstring injuries into account.¹¹⁷ Clinicians aiming to design evidence-based hamstring injury prevention programs for athletes are therefore advised to be cautious about how findings from risk factor studies are interpreted.

To further understand the role that hamstring strength has in creating injury risk, a prospective study has recently been completed investigating whether or not eccentric hamstring strength, measured during the Nordic hamstring exercise, is a risk factor for hamstring strain in elite Australian rules football players.¹¹⁸

In this study, a novel field-testing device, the NordBord (Fig. 34.19), was used to record eccentric strength while players performed the Nordic hamstring exercise. Force was recorded by connecting uniaxial load cells between the braces holding the ankles and the base of the testing device, allowing the force output for each lower limb to be recorded independently. Peak force during the exercise from a single lower limb averaged 330 N for



Figure 34.19 The NordBord test

uninjured football players. Players with a peak force of less than 279 N at the end of the pre-season were found to have a 4.3 fold greater risk of sustaining a hamstring strain during the season. Similar results have also recently been reported in elite soccer players.¹¹⁹ Taken together, these findings suggest that athletes with low levels of eccentric hamstring strength are at greater risk of injury.

Musculotendon morphology

While the available evidence is somewhat limited at this stage, intrinsic musculotendon morphological properties may have a role in determining an individual's risk for hamstring strain. For example, the width of the proximal biceps femoris long head tendon exhibits high variability among healthy athletes,¹²⁰ and tissue strains in the muscle fibres immediately adjacent to the proximal musculotendinous junction have been found to be higher in individuals with a narrow proximal tendon width.^{121, 122} Thus, it is possible that athletes with a narrow proximal tendon have an increased risk for proximal biceps femoris long head strains.

Smaller muscle fascicle length might be another morphological property associated with greater risk for injury, as it has been recently reported that athletes with a history of unilateral hamstring strain have reduced muscle fascicle length in the biceps femoris long head on the injured side compared to the contralateral uninjured side and compared to people without a history of injury.¹¹⁴ Furthermore, a recent prospective study has found that risk of hamstring injury is significantly greater for soccer players with biceps femoris long head fascicle lengths shorter than 10.56 cm.¹¹⁹

Other factors

Other intrinsic risk factors for acute hamstring strain which have only limited supporting evidence include (but are not limited to): ethnicity,^{23, 105} reduced ankle

dorsiflexion lunge range of motion,¹²³ reduced quadriceps flexibility^{91, 123} and poor lower-limb joint position sense.⁸⁶

There is conflicting evidence regarding whether or not reduced hamstring flexibility (e.g. assessed via the active and/or passive knee extension test) is a risk factor for hamstring strain. Most studies indicate that reduced hamstring flexibility is not a risk factor;^{37, 102, 103, 123} however, two studies have reported the contrary.^{91, 124}

Extrinsic risk factors

Fatigue

It has long been speculated that fatigue is a risk factor for an acute hamstring strain, but there is very little evidence to support or refute this claim. Verrall et al. found that 85% of acute hamstring strains occurred after the first quarter of a competitive match or after the first 15 minutes of a training session.⁸ Furthermore, Woods et al. found that 47% of their acute hamstring strains occurred towards the end (during the final third) of the first and second halves of the match.²³ Such observations suggest fatigue may be a factor, but further research is required.

Player position

There is limited evidence that different playing positions are associated with a higher risk of hamstring strain. Goalkeepers have a significantly lower risk for hamstring strain than outfield players in soccer²³ and rugby forwards have a reduced risk of hamstring strains compared to backs.³⁶ In American football, the speed position players, such as the wide receivers and defensive secondary, as well as players on the special teams units, are most commonly injured.²⁰

PREVENTION OF HAMSTRING STRAINS

A systematic review published in 2010 concluded that there is insufficient evidence from randomised controlled trials to draw conclusions on the effectiveness of interventions used to prevent hamstring injuries in people participating in football or other high-risk activities for these injuries.¹²⁵ However, recent studies have provided strong evidence indicating that eccentric hamstring strength training with good compliance is an effective intervention for preventing hamstring strains.¹²⁶

Eccentric hamstring strength training

High-quality studies have demonstrated that eccentric strength training for the hamstring muscles can prevent hamstring strains in athletes.^{75–77, 127}

In Petersen's study,⁷⁶ there was a significant reduction (approximately 3-fold) in the 'total' number of hamstring injuries (i.e. new plus recurrent injuries) and a significant

reduction (approximately 7-fold) in the number of recurrent injuries for the intervention group, which undertook a 10-week progressive pre-season eccentric training program of Nordic exercises (Table 34.4) followed by a weekly seasonal program. There was also an approximate 2.5-fold reduction in the number of new hamstring injuries in the intervention group, but this effect did not reach statistical significance.

Arnason's study also showed a reduction in hamstring injuries with a Nordic exercise program,⁷⁵ while a positive effect on prevention of hamstring injuries in soccer players was also found for an eccentric/concentric strengthening program using a Yo-Yo flywheel ergometer.¹²⁶ Most recently, van der Horst et al. investigated the preventive effect of the Nordic hamstring exercise in amateur soccer players and found a significant reduction in the incidence of hamstring strains but not the severity.⁷⁷

In contrast to the aforementioned studies, two other studies have investigated the effectiveness of the Nordic hamstring exercise. Gabbe et al.¹²⁸ reported a minimal effect only, while Engebretsen et al.¹²⁹ reported no effect. However, both of these studies suffered from poor compliance, which most likely explains the less favourable results.¹²⁵

Balance exercises/proprioception training

Proprioceptive exercises or balance training may be an effective strategy for preventing hamstring injuries.^{130–132} A positive effect was found in one study,¹³⁰ while two other studies failed to show any effect.^{131, 132}

In the German study, 24 elite female soccer players of a premier league soccer team included an additional soccer-specific proprioceptive multistation training program over 3 years.¹³⁰ Progression in level of difficulty from easy to complex was a main feature of the exercises. The duration of each exercise was between 15 and 30 seconds.

The following exercises were implemented:

1. single-leg stand on right and left foot
2. jump forwards in single-leg stand with flexed knee at landing and balancing
3. jump backwards in single-leg stand and balancing
4. row jumping single foot
5. row jumping bipedal
6. obstacle course forwards and backwards
7. obstacle course sideways
8. bipedal jumping on forefoot
9. sideways jumping in single-leg stand
10. sitting on a wobble board with balancing torso
11. jumping forwards over a line, landing with flexed knees and balancing
12. standing on both hands and feet with diagonal balancing.

All exercises were performed with no additional weight; that is, players had to bear only their own body weight on one or two legs or all extremities depending on the exercise. In addition, balance training was implemented in soccer-specific match play training on balance boards.

At the end of the 3-year proprioceptive balance training intervention, non-contact hamstring injury rates were reduced from 22.4 to 8.2/1000 hours. Furthermore, the more minutes of balance training performed, the lower the rate of hamstring injuries.

Sport-specific training

In seasonal sports such as Australian rules football and soccer, hamstring strains occur far more frequently during official matches/games than training sessions. Hence, sport-specific training during the pre-season is considered to be an important strategy for preventing hamstring strains during the competitive season.¹³³

The aim of sport-specific training during the pre-season is to gradually expose the hamstring muscles to the sorts of high-risk activities that are regularly encountered during the competitive season (e.g. repetitive sprinting, maximum accelerations, kicking, running with trunk flexion), thereby improving the ability of the hamstrings to withstand these activities. For instance, Verrall et al. demonstrated a significant reduction in the incidence of hamstring strains in one Australian rules football club following the introduction of sprint training, stretching once fatigued and sport-specific training drills during the pre-season.⁸⁷

A promising clinical approach for the high-risk athlete

Another potential strategy for preventing hamstring strains, especially for the athlete with a past history of injury, is to try to identify (via clinical examination) those who have reacted adversely to a high-intensity activity (e.g. match/game or training session). In some instances, high-intensity activities may result in aberrant levels of muscle damage and/or fatigue. Athletes can be assessed 1–2 days following a match/game throughout the competitive season and high-risk activities can be modified accordingly for those who display positive clinical signs (e.g. pain and inhibition with resisted muscle contraction and overt tenderness with palpation).

Isometric maximum voluntary contractions may be most suitable for this purpose because they are quick and easy to perform for the clinician, no expensive equipment is necessarily required and they are relatively safe for the athlete. Recent research has demonstrated the potential capability of isometric maximum voluntary contractions of the hamstrings to detect the presence of

neuromuscular inhibition for up to 72 hours following a soccer match.^{134, 135}

A simple, clinically applicable approach for measuring isometric maximum voluntary contractions has been described by Schache et al.¹³⁶ A digital sphygmomanometer can be used as an inexpensive way to quantify the athlete's hamstring strength. Alternatively, a portable force plate can be used to obtain data in newtons (N).¹³⁵ Data can be collected in different positions that change the way load is distributed on the hamstrings; for example, with the athlete positioned with their knee flexed to 90° and heel resting on a firm surface (Fig. 34.20a) or with the athlete positioned in supine with their knee extended and heel resting on a small step performing a single-leg bridge (Fig. 34.20b). Both of these positions mimic commonly used pain provocation tests for the hamstrings. Other clinically applicable and reliable methods for recording isometric maximum voluntary contractions of the hamstrings involve the use of a hand-held dynamometer or load cell and are performed in various prone or seated positions against either therapist resistance or external belt fixation.^{92, 137, 138}

The overall objective is to identify those athletes who have painful and/or inhibited hamstring muscles. It is then up to the clinician to use these findings together with the athlete's history and other clinical examination findings to decide whether or not an intervention is required, and if so, what type of intervention (e.g. manual therapy, specific exercise therapy, activity modification). Clinically applicable tests of hamstring strength are also very useful outcome measures for monitoring healing following acute hamstring strains.³⁴



(a)

Figure 34.20 Schache's hamstring maximum voluntary contraction (MVC) test.¹³⁶ The set-up used to measure an isometric MVC of the hamstrings with the digital sphygmomanometer (a) The athlete is positioned with his or her knee flexed to 90° and heel resting on a firm surface



(b)

Figure 34.20 (cont.) (b) The athlete positioned in supine with their knee extended and heel resting on a small step performing a single-leg bridge

REFERRED PAIN TO POSTERIOR THIGH

The possibility of referred pain should always be considered in the athlete presenting with posterior thigh pain. Hamstring pain may be referred from the lumbar spine, the SIJ or from soft tissues; for example, the proximal fibres of the gluteus maximus, and especially the gluteus medius and the piriformis muscle. Often, there is a history of previous or current low back pain.

The slump test (Fig. 34.2g) should be used to detect neural tightness. The test is positive when the patient's hamstring pain is reproduced and subsequently relieved with reduction of the neural tension by neck extension. Examination may reveal reduced range of movement of the lumbar spine, tenderness and/or stiffness of lumbar intervertebral joint(s) or tenderness over the area of the SIJ.

A positive slump test is strongly suggestive of a referred component to the patient's pain. The slump test has been advocated as a method of treatment of hamstring pain in Australian rules football players.¹³⁹ Nevertheless, a negative slump test does not exclude the possibility of referred pain and the lumbar spine should be carefully examined to detect any intervertebral segment hypomobility. Careful palpation of the hip musculature, especially the extensors, abductors and external rotators, should also be performed.

Trigger points

Trigger points are common sources of referred pain to both the buttock (Chapter 30) and posterior thigh. The most common trigger points that refer pain to the

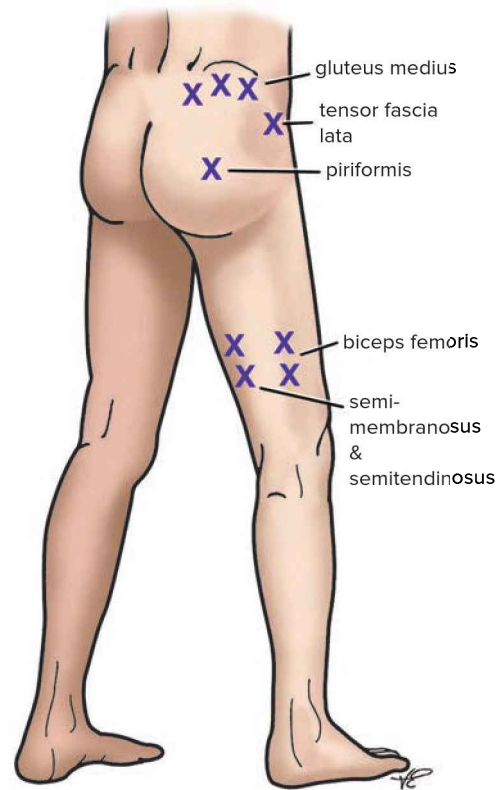


Figure 34.21 Pattern of referred pain to the hamstrings from trigger points

mid-hamstring are in the gluteus minimus, gluteus medius and piriformis muscles (Fig. 34.21).

The clinical syndrome associated with posterior thigh pain without evidence of hamstring muscle injury on MRI and reproduction of the patient's pain on palpation of gluteal trigger points is now well recognised and extremely common.¹⁴⁰

The clinical features are described earlier in Table 34.2. The patient will often complain of a feeling of tightness, cramping, 'twinge' or a feeling that the hamstring is 'about to tear'. On examination there may be some localised tenderness in the hamstring, although it is usually not focal and there is restriction in hamstring and gluteal stretch. Firm palpation of the gluteal muscles will detect tight bands that contain active trigger points, which when firmly palpated are extremely tender, refer pain into the hamstring and may elicit a 'twitch response'.

Treatment involves deactivating the trigger point either with soft tissue massage techniques (Fig. 34.22a) or dry needling (Fig. 34.22b). Following the local treatment, the tight muscle groups (the gluteals and hamstrings) should be stretched.



(a)



(b)

Figure 34.22 Treatment of gluteal trigger points (a) Elbow pressure (b) Dry needling

Lumbar spine

The lumbar spine is a source of pain referral to the posterior thigh. Unfortunately, it is difficult to distinguish between sources based on the behaviour and distribution of the pain. Pain may be referred from the disc, zygapophyseal joints, muscles, ligaments or any structure that can produce pain locally in the lumbar spine.¹⁴¹

Nerve root compression may also be a cause of hamstring pain. Diagnostic blocks and provocation injections have been advocated to isolate sources of pain in the lumbar spine. However, in the clinical setting, this is often not possible. It is important to examine the lumbar spine carefully (Chapter 29). This will assist in the identification of the lumbar spine as a source of hamstring pain. It is also important to remember that the lumbar spine may be a cause of lumbar pain indirectly. For example, the lumbar spine may cause a biomechanical block to hip extension, resulting in overload of the SIJ and referred pain to the hamstring group.

True nerve root compression is usually more definitive in its presentation. The patient may have associated neurological symptoms, such as numbness and loss of foot

eversion. The management of these injuries usually involves an extended period of rest and, in certain cases, an epidural injection. In extreme cases, surgical decompression of the nerve root may be warranted.

Spondylolisthesis and spondylolysis (Chapter 29) have both been associated with hamstring pain and tightness.¹⁴² Examination findings of positive lumbar quadrant tests or single-leg standing lumbar extension suggests either of these conditions and these spinal pathologies can be confirmed on MRI or CT scan.

Stabilisation programs are the treatment of choice as the deep abdominal muscles are deficient in people with back pain as a result of spondylolisthesis and spondylolysis.¹⁴³ In severe cases, clinicians have resorted to corticosteroid injection (+/- neuroablation using pulsed radiofrequency) under X-ray control into the deficient pars interarticularis.¹⁴⁴ This procedure is only supported by low-level evidence, but it may reduce pain from spondylolysis.

Not all hamstring pain associated with the lumbar spine is due to sport-related loading of the lumbar spine. The lumbar spine can cause pain as a result of prolonged sitting

or bending forwards. Athletes in sedentary occupations should be aware that poor sitting posture can aggravate lumbar-mediated posterior thigh pain.

Travel involving prolonged sitting prior to training and competition may cause injury. Prolonged sitting results in sustained lumbar flexion whereas running requires good lumbar extension. When the lumbar spine is required to suddenly 'switch' from one activity to the other, this may cause problems. Care should be taken to limit prolonged sitting and to provide adequate lumbar support.

Sacroiliac complex

SIJ abnormalities can refer pain into the hamstring or cause indirect pain in the hamstring similar to the lumbar spine. Problems of the SIJ are discussed in Chapter 30.

OTHER HAMSTRING INJURIES

Avulsion of the hamstring from the ischial tuberosity

with RAJ SUBBU and FARES HADDAD

The proximal hamstring complex is frequently injured in athletes, with 12% of hamstring injuries being in the proximal complex and 9% being complete ruptures.¹⁴⁵ Traditionally, these occur in waterskiing, sprinting and hurdling; however, they are increasingly being recognised in a variety of different sports, commonly soccer and rugby.¹⁴⁶ The injuries are typically associated with a very forceful eccentric contraction of the hamstrings where the hip is rapidly flexed with the knee in full extension.

Patients present with a history of hearing a pop or snap at time of injury, complaining of acute pain at the buttock crease, an antalgic gait, and loss of hip and knee function with reduced hamstring power. Physical examination may reveal swelling and massive ecchymosis in the posterior aspect of the thigh, which may even track distally. There may also be a positive bowstring sign when the patient is assessed in the prone position with a palpable lack of tension on passive flexion of the knee to 90°. Palpation at the gluteal crease evokes pain and there is decreased power in both flexion of the knee and extension of the hip. Tenderness proximally with symptoms and signs of sciatic nerve irritation should alert the clinician to the possibility of an avulsion injury.

Once clinically assessed, radiological imaging is required to confirm the diagnosis. MRI and ultrasonography remain the imaging modalities of choice (Fig 34.23).¹⁴⁵ With greater severity of injury there is a greater amount of haemorrhage, oedema and fluid involving a larger longitudinal and cross-sectional area.¹⁴⁵ Studies have compared MRI and ultrasonography, and MRI has been shown to consistently demonstrate large areas of oedema and greater sensitivity

in picking up subtle fluid changes and exact anatomical area of injury.¹⁴⁷ A specific MRI classification for proximal avulsion injuries has been described based on the anatomical location and amount of retraction of the proximal complex (Table 34.5).¹⁴⁷ Typically, type 5 injuries are delayed presentations, indicating that the tendons may retract with time (Fig 34.23b).

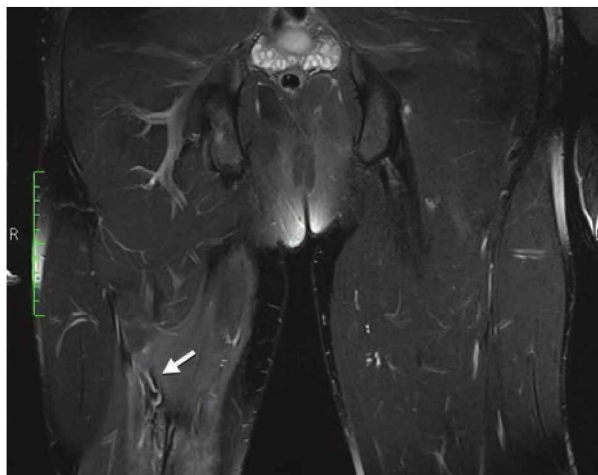
Treatment has previously been based on the amount of retraction from the ischial tuberosity with less than 2 cm being treated conservatively, which may lead to decreased hamstring strength, function and prolonged rehabilitation.¹⁴⁸ A recent study of 112 proximal avulsion injuries has shown that surgical repair is indicated in all proximal avulsion injuries despite the amount of retraction.¹⁴⁶

The timing of surgery is important regarding RTP. Subbu et al. reported that the average time to RTP for those with surgical repair within 6 weeks of initial injury was 16 weeks (range 12–32), 25 weeks (range 18–40) within 6 months and 29 weeks (24–41) after 6 months of initial injury.¹⁴⁶ Return to full pre-injury level of sport was on average 9 weeks faster for those with surgical repair within 6 weeks of injury compared with those within 6 months, and 13 weeks faster compared with those with surgical repair performed more than 6 months after injury.¹⁴⁶



(a)

Figure 34.23 (a) MRI showing acute complete avulsion of the conjoint tendon of semitendinosus and biceps femoris with retraction (red arrow). Semimembranosus intact (white arrow)



(b)

Figure 34.23 (cont.) (b) MRI showing chronic complete avulsion of the entire proximal hamstring complex with diffuse increased signal and early muscle atrophy. Ischial tuberosity fragment identified (arrow). Consistent with type 5b injury

The surgical procedure involves retrieving the retracted complex with suture anchors that have a metal peg, which are fastened into the ischial tuberosity. The synthetic sutures attached to the metal peg can then be passed through the mobilised tendon to fashion a well-tensioned bone-tendon approximation. When surgery is delayed, the procedure becomes more complex with greater retraction of the proximal complex; a larger, vertical incision is required for greater exposure and there is an increased risk of damage to the sciatic nerve, with it becoming embedded within the retracted complex. When surgery is delayed until after 6 weeks, patients inevitably require postoperative bracing which consequently delays the rehabilitation process.

A structured rehabilitation program should be centred on using the first 4 weeks as healing time while protecting the surgical repair. The next 4 weeks are used to regain range of motion and function and then a graduated return to sport-specific activity.¹⁴⁹ Each patient should be given an individualised rehabilitation program depending on the complexity of surgery and the level of activity to be achieved. A rehabilitation guide is outlined in Table 34.6.

Early diagnosis and surgical interventions are associated with less complex surgery, fewer postoperative complications, more efficient rehabilitation and a faster return to pre-injury levels of activity.

Upper hamstring tendinopathy

Tendinopathy of the proximal hamstring tendon or 'high hamstring tendinopathy' is increasingly being recognised as an important cause for chronic pain in the active population (Chapter 30). Patients usually present with subacute onset of deep buttock or posterior thigh pain/stiffness that is exacerbated by repetitive activity (such as long-distance running) and is often aggravated by sitting, stretching and aggressive strengthening. MRI is more sensitive than ultrasound in detecting pathological abnormalities involving the proximal hamstring tendons, but note that many typical findings such as intratendinous and peritendinous signal as well as ischial tuberosity oedema can also be seen in asymptomatic individuals.

Initial treatment in clinical practice should include a graduated strengthening program, largely comprising isometric and slow isotonic exercises under high load. However, patients sometimes fail to respond to this conservative regimen. The use of ultrasound-guided injections of corticosteroid and local anaesthetic has been shown to be safe and effective in reducing pain in the short term, but is associated with a high recurrence rate when used in isolation.¹⁵⁰ This procedure should be performed in conjunction with a progressive strengthening program to achieve optimal results and avoid recurrent injury.

Table 34.5 MRI classification of proximal avulsion injuries¹⁴⁷

Type	Level of injury
Type 1	Osseous avulsions (generally seen in paediatric patients)
Type 2	Avulsions at the musculotendinous junction
Type 3	Incomplete tendon avulsions
Type 4	Complete avulsions with no/minimal retraction
Type 5a	Complete avulsions with retraction but NO sciatic nerve involvement
Type 5b	Complete avulsions with retraction AND sciatic nerve involvement

Table 34.6 Rehabilitation protocol for proximal avulsion injuries

Weeks	Rehabilitation guide
1–2	Avoid hip flexion beyond 60° Wound care Avoid sitting on affected ischial tuberosity Calf pumps Non weight-bearing
3–4	Start passive knee flexion and hip extension
5–6	Bear toe weight if possible
7–10	Encourage to bear full weight Hydrotherapy Passive and active range of motion—avoid extremes Begin core work Begin closed chain exercises
11–14	Expect normal gait Encourage strength work Progress to fast walk/jog
15–16	Consider isokinetic testing Heavy weight training Potentially running

A rehabilitation program utilising eccentric training, lumbopelvic stabilisation and trigger point dry needling has been suggested.¹⁵¹ Shock wave treatment has also been shown to be effective in relieving symptoms associated with high hamstring tendinopathy.¹⁵²

In order to document the severity of the tendinopathy, the VISA-H questionnaire has recently been developed.¹⁵³

LESS COMMON CAUSES

Nerve entrapments

The hamstring group is supplied by the tibial branch of the sciatic nerve except for the short head of the biceps femoris, which is supplied by the peroneal branch of the sciatic nerve. These nerves arise from the lumbosacral plexus, specifically from the roots of L5, S1 and S2.

Nerve damage can occur at a variety of sites, resulting in pain in the posterior thigh. Compression of the nerve roots of the sacral plexus will often result in pain into the hamstring group. Usually this is distinguished from other conditions by the identification of associated neurological symptoms such as alteration in sensation, loss of the Achilles reflex or weakness in muscles not in the hamstring group, such as the ankle evertors.

The sciatic nerve may be damaged or compressed at any point along its pathway as a result of direct impact or pelvic trauma. Compression of the nerve at the level of

the piriformis has been described as an alternative cause of sciatica.¹⁵⁴ However, work by McCrory et al. suggests that it is not only the piriformis, but also the other hip external rotators that may cause compression of the sciatic nerve.¹⁵⁵

The peripheral nerves may also be a source of posterior thigh pain. The posterior cutaneous nerve of the thigh (PCNT) and the inferior cluneal nerve supply the skin over the posterior thigh. The PCNT has been described as the source of pain in piriformis syndrome as an alternative to the sciatic nerve.¹⁵⁵ If symptoms do not extend below the knee and there is no associated loss of neurological function in the structures supplied by the sciatic nerve, then the PCNT should be considered.

Ischial bursitis

It is often difficult to distinguish between high hamstring tendinopathy and ischial bursitis. Both conditions present as pain at the origin of the hamstring muscle. An inflamed bursa is not readily palpated; however, athletes tend to complain of pain when sitting on hard surfaces where the ischium is under pressure. Ultrasound or MRI can confirm the presence of a fluid-filled bursa.

Anti-inflammatory medications combined with ice and rest are of limited benefit. Corticosteroid injections can be performed under radiographic control but the results are not always satisfactory.

Adductor magnus strains

Adductor magnus strains are rare, but when they do occur, behave similarly to a hamstring strain. The mechanism tends to be more of a rotatory action of eccentric internal rotation on one hip or landing with sudden deep hip flexion. Prognosis tends to be far better than hamstring strains; therefore, it is important to differentiate it from strains in the hamstring. The key to differentiating this condition is careful palpation to elicit the precise location of the tissue damage. Side-lying on the affected leg allows that hamstring group to fall laterally so that ready access can be made to the adductor magnus.

Compartment syndrome of the posterior thigh

Although nowhere near as common as lower leg compartment syndromes, the posterior thigh can be affected by a compartment syndrome. Patients present with dull pain, stiffness, cramps and weakness of the posterior thigh during and after training.¹⁵⁶ Two groups of patients with this syndrome are seen—endurance athletes without a history of trauma and those with a history of hamstring injury. Conservative management has not been successful and posterior fasciotomy of the thigh appears to be an effective treatment.¹⁵⁶

Vascular

Endofibrosis of the external iliac artery usually produces pain in the lateral and anterior thigh or calf (Chapter 39). However, in some cases pain may be experienced in the posterior thigh. This condition is associated with cycling and has been observed in triathletes.¹⁵⁷ The pain is claudicant in nature. Pain may arise after 15–20 minutes of

exercise but usually stops immediately with the cessation of exercise.

On examination, a bruit is heard during the exercise that causes the pain. Diagnosis may be confirmed with echography or arteriography. If the condition is affecting performance, then surgical treatment is indicated.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e